Colombia

A business case for sustainable coffee production

SUSTAINABLE COFFEE PROGRAM

August 2014

Executive summary

Colombia is the world's second largest Arabica producer, accounting for 15 percent of global Arabica exports. Colombia has a rich tradition of coffee production, an efficient supply chain, and a premium market for most of its coffee exports. Coffee farming remains the leading source of livelihood for rural Colombians, though its importance to the broader economy has declined in significance over the past several decades.

95% of Colombian farms have less than 5 hectares of coffee and account for over 70% of national production. Between 2008 and 2012, Colombia's output dropped due to adverse weather conditions and a national tree renovation program to combat coffee leaf rust. Estimates suggest that over 50% of coffee areas have been renovated so far. As these new trees start bearing, production has started to rise again and amounted to 10.9 million bags in 2013.

Farmers' margins from coffee have been eroding. The costs associated with key production inputs including labor and fertilizer have increased rapidly over the past decade. Rising costs have made Colombia less competitive than other Arabica origins. In 2013, for the first time the Colombian government provided direct cash subsidies to farmers and paid more than US$ 600 million to bolster farmers' incomes. As a result, farmers are increasingly weighing the opportunity cost of coffee farming, both in terms of land and labor (time).

Colombia's coffee sector is at a crossroads. Efforts targeted at producers should focus on making farms more profitable. No single solution can address this challenge. Some farmers have opportunities to excel in specialty coffee; others may need to diversify their farm enterprises beyond coffee. However, for a large subset of the farmer population, there exist opportunities to boost yields, improve labor productivity, and reduce overall farming costs. These interventions offer the combined potential to increase farmers' net incomes by 50% and reduce the overall amount of family labor required for coffee farming.

If this issue is not addressed, it will be difficult for certification and verification to expand. Colombia was an early adopter of certification/ verification and has a higher share of certified/verified sales than other countries surveyed by IDH. In addition, large volumes are already double-certified. However, economic constraints may limit future growth. With farmers already facing high production costs, there is resistance to the creation of new costs in the supply chain. New environmental laws, particularly regarding wastewater disposal, are likely to make compliance more difficult and costly for individual small farms in the future. Improving the profitability of farming would make it easier for farmers to absorb added sustainability costs and thus improve the business case for them to invest in sustainability.

There is a convincing business case for private and public market actors and the government to co-invest in these efforts in order to improve the competitiveness of the sector and to avoid the exit of coffee farmers. An ambitious program could reach potentially up to 50% of active farmers (230,000 farmers) over 10 years and increase production by 1.5 million bags. This would raise coffee export revenues by US$ 260 million while achieving cost savings of US$ 75 million. Implementing these opportunities would requiring new approaches for reaching farmers and cost the sector approximately US$ 200 million over 10 years.

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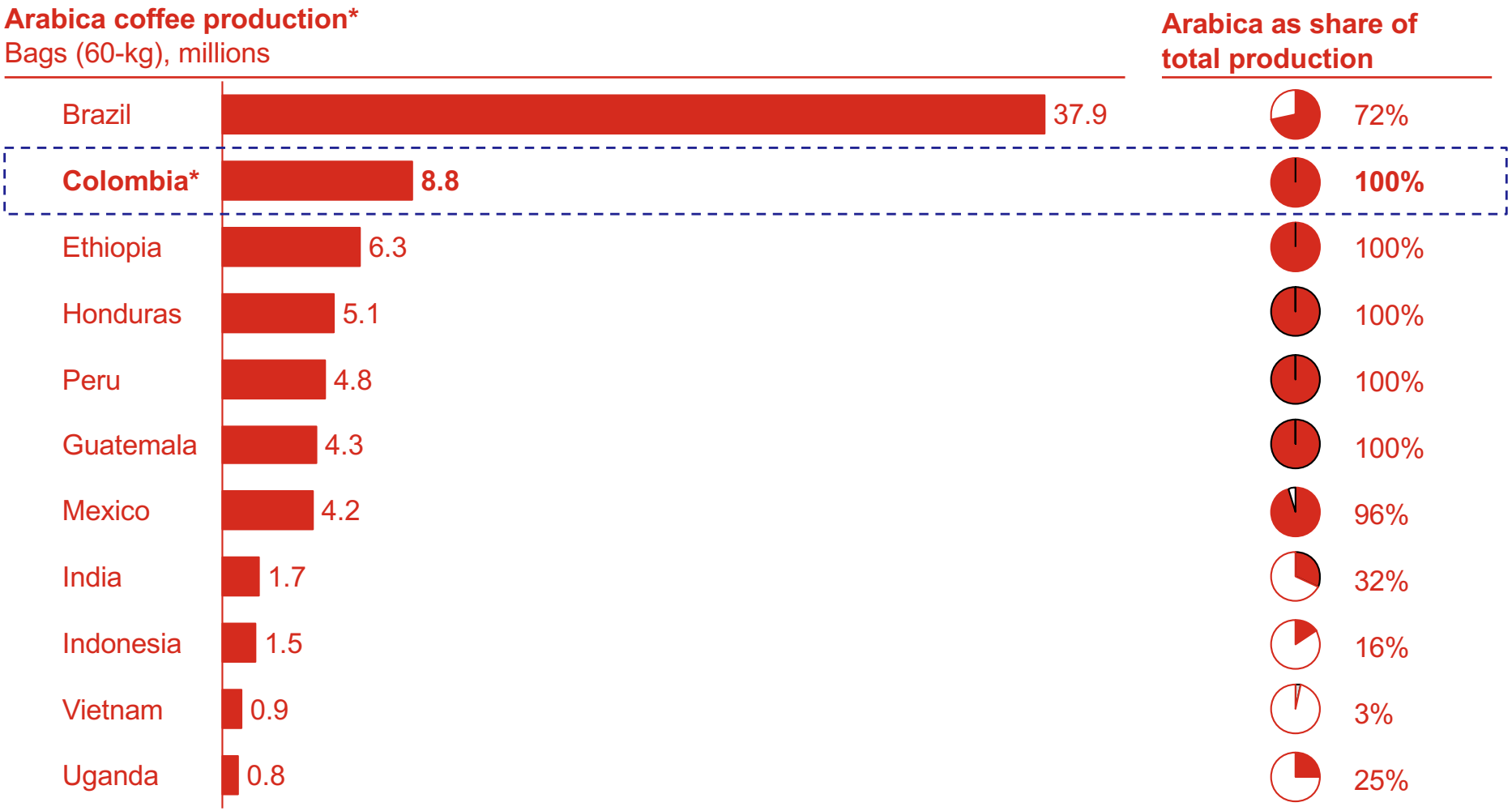
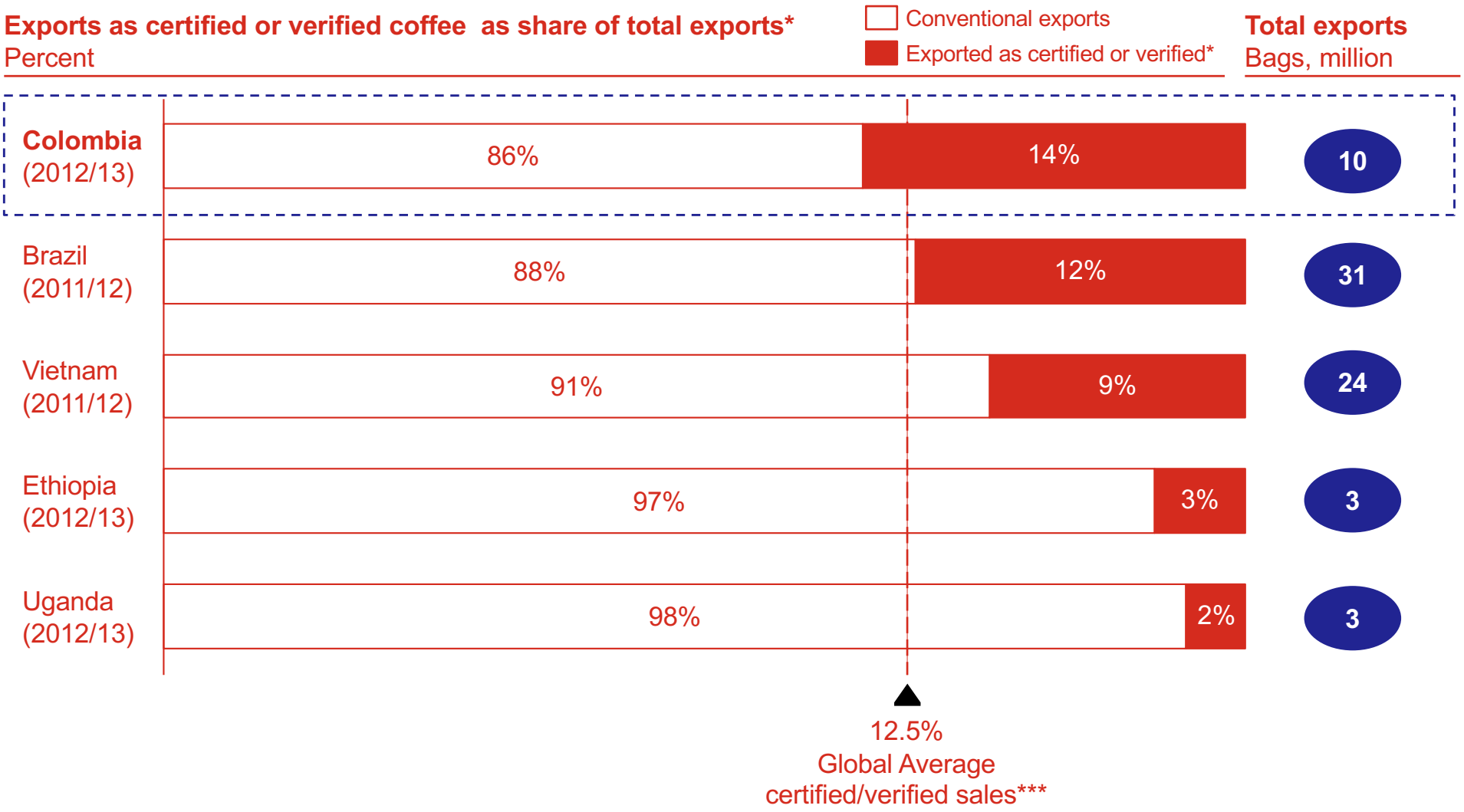
- **Colombia coffee industry primer**
- Analytical framework
- Colombia business case discussion

Colombia is the world's second largest Arabica producer



* Average of 2011/12 and 2012/13 coffee years
Source: USDA; TechnoServe analysis

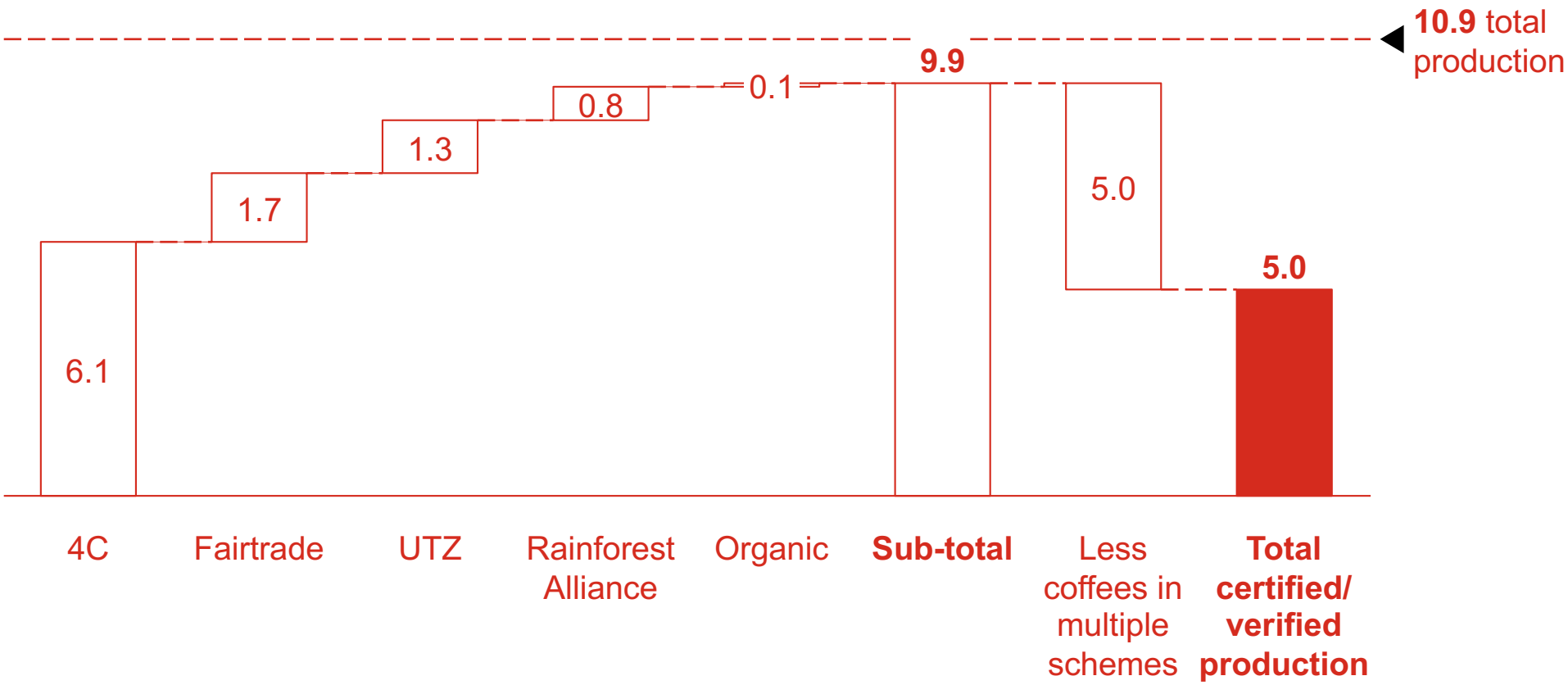
About 14% of Colombia's exports are either certified or verified



* Most recent year of available data
** Based on 2013 data; Fairtrade data not available at time of publication; excludes Nespresso AAA and Starbucks C.A.F.E. Practices
Source: USDA; ICO; TCC Coffee Barometer 2012; stakeholder interviews

Roughly 45% of production is under one or more certification or verification

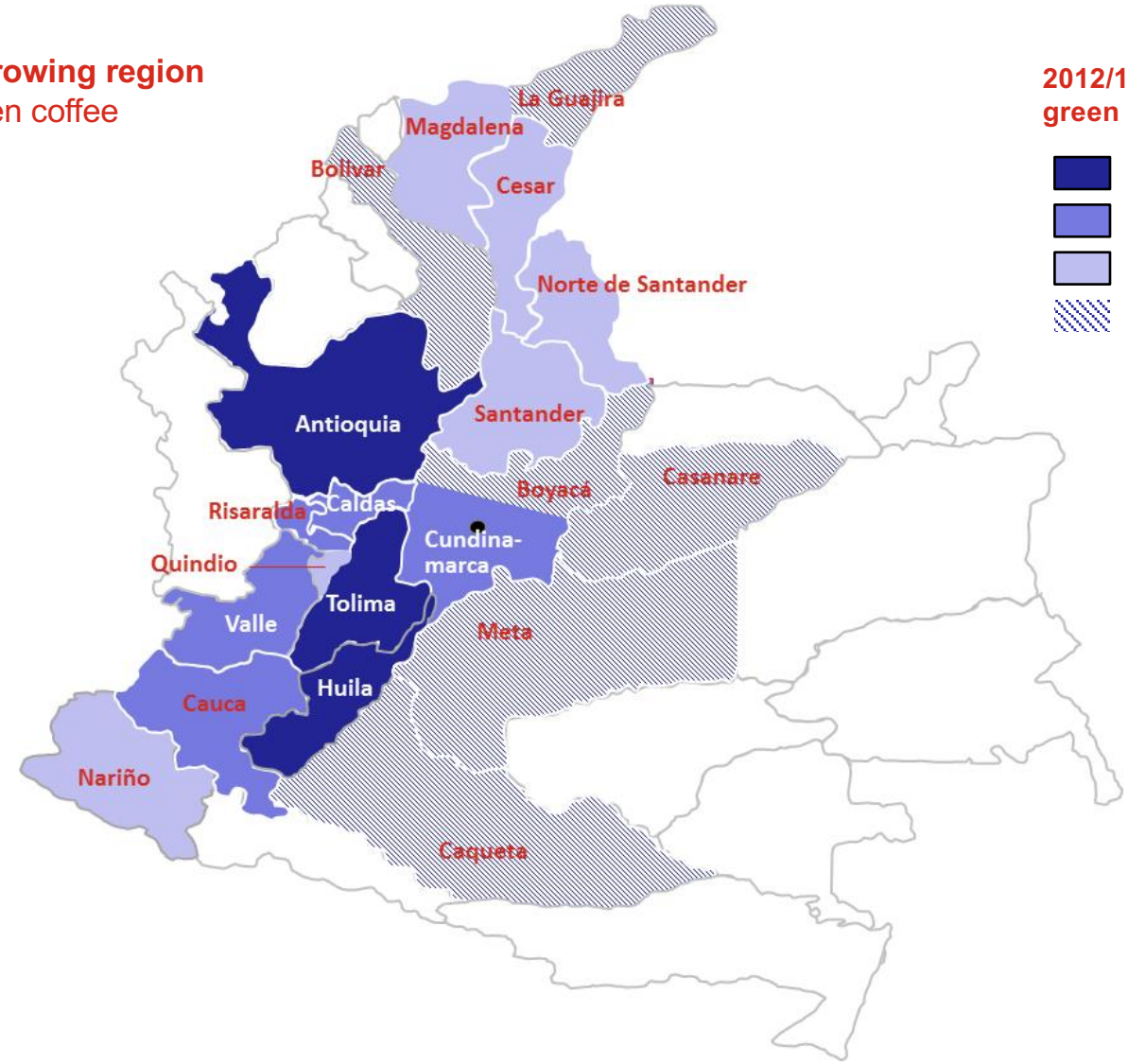
Coffee production by certification/verification, 2013
Bags (60-kg), millions



Source: Stakeholder interviews; FNC; TechnoServe analysis

Colombia has a range of Arabica growing regions and more than half a million farms

Production per coffee growing region
2012/13, million bags green coffee



2012/13 production of bags green coffee

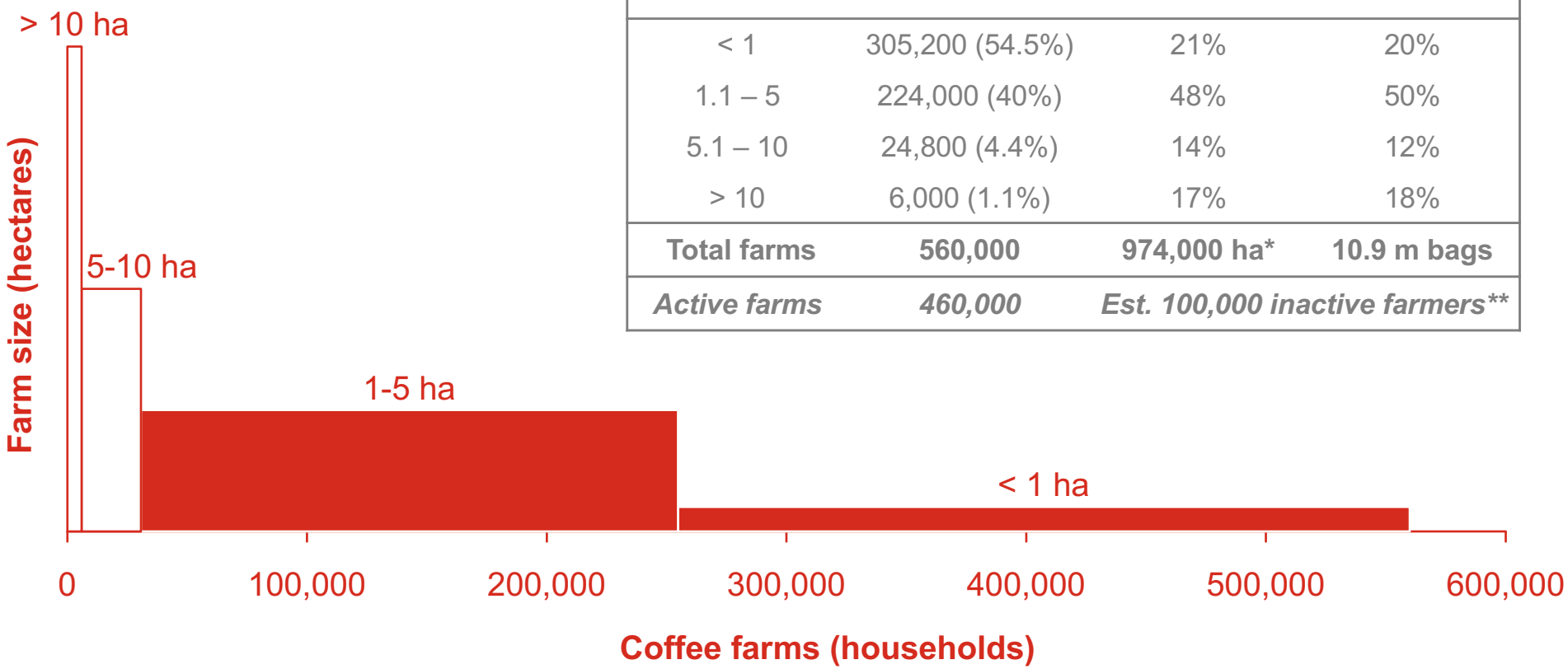
- > 1 million bags
- 0.5 – 1 million bags
- 0.1 – 0.5 million bags
- < 0.1 million bags

Source: Agronet, FNC

95% of Colombian coffee farms have less than 5 ha of coffee

Distribution of Colombian coffee farms by coffee farm size

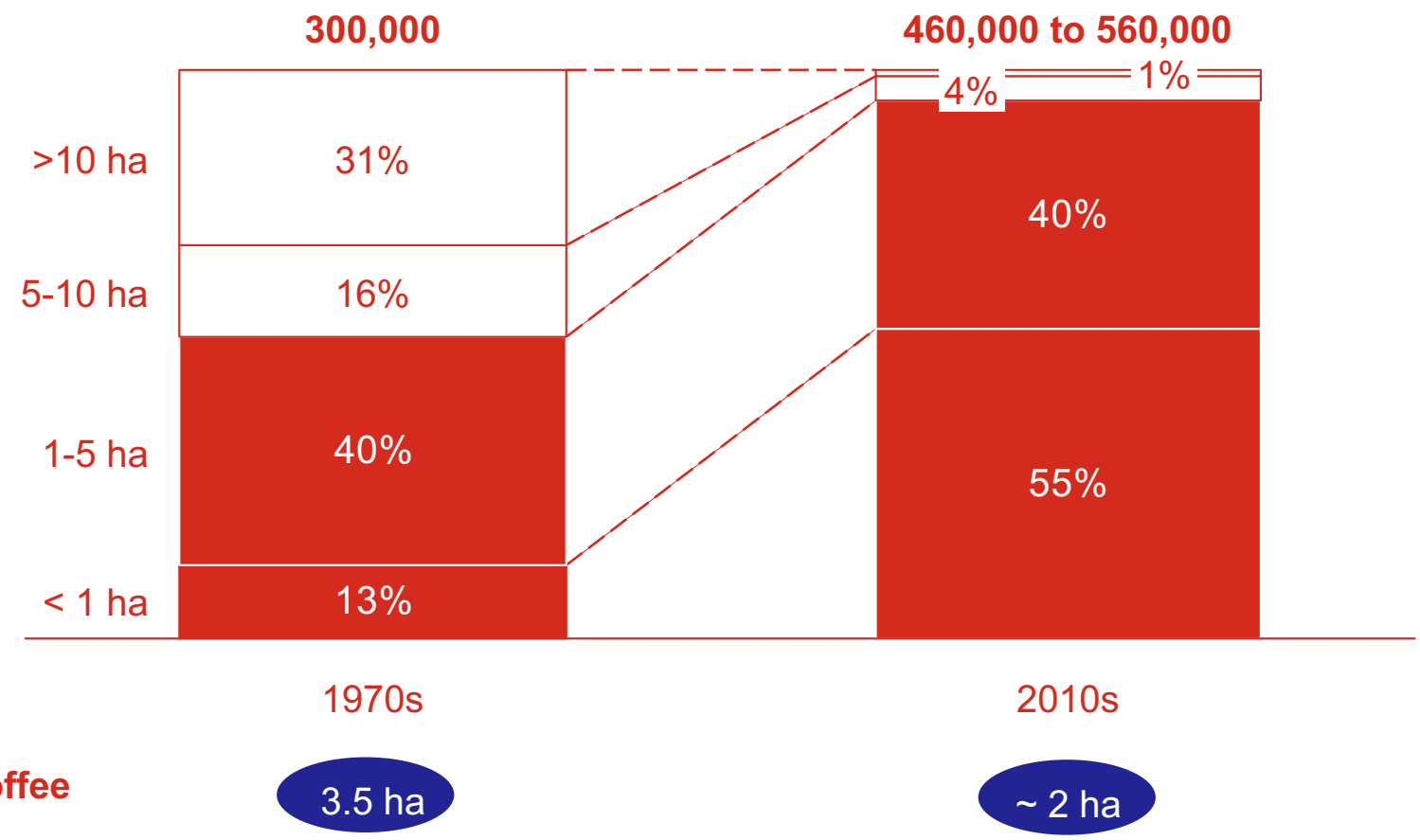
Area under coffee (ha)	Number of coffee farms	Share of coffee area	Share of production
< 1	305,200 (54.5%)	21%	20%
1.1 – 5	224,000 (40%)	48%	50%
5.1 – 10	24,800 (4.4%)	14%	12%
> 10	6,000 (1.1%)	17%	18%
Total farms	**560,000**	**974,000 ha***	**10.9 m bags**
Active farms	***460,000***	***Est. 100,000 inactive farmers****	



* Based on planted coffee area
** > 100,000 farmers did not take advantage of the direct income subsidy paid by the government (PIC scheme)
Source: FNC; interviews

A generation ago, only half of Colombia's farms were smaller than 5 hectares

Coffee farms by farm size
Percentage of coffee farms



Note: Data based on 1970 National Coffee Census and 2013 SICA data
Source: FNC; FONAC

A combination of demographic and economic factors have contributed to diminishing farm size

Decreasing farm size due to inheritance traditions

- Traditionally, families distributed coffee land between heirs
- With average household sizes of up to 6 this led to considerably decreasing farm sizes

Exit of large farms due to unfavorable economics

- High cost structure due to large share of external farm workers has forced larger farms to give up their farms, in particular in times of low coffee prices (e.g., beginning of the 90s)
- Smaller farms have proven to be more resistant to price falls due to lower cash costs

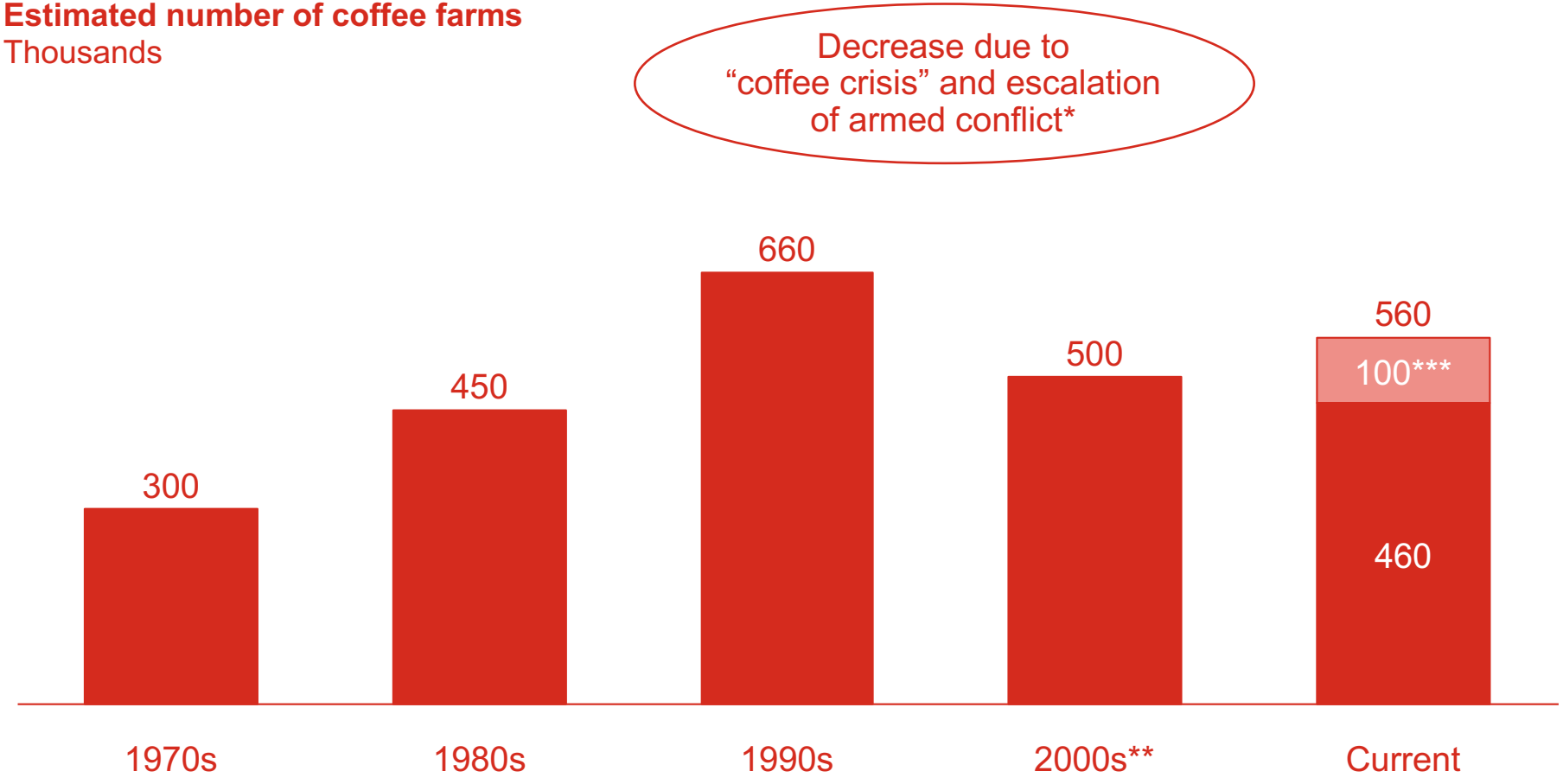
Many new small farms due to attractiveness of sector

- Many farmers moved into the sector, starting to grow coffee on small farms because they perceived coffee to be an attractive and stable income alternative
- Coffee sector is only sector with purchase guarantee and high institutional support (e.g. technical assistance, transfer of technological innovation, credit, infrastructure, social protection, health services, education programs)

Source: World Bank, stakeholder interviews

The number of coffee farms peaked in the 1990s, but fell in the early 2000s

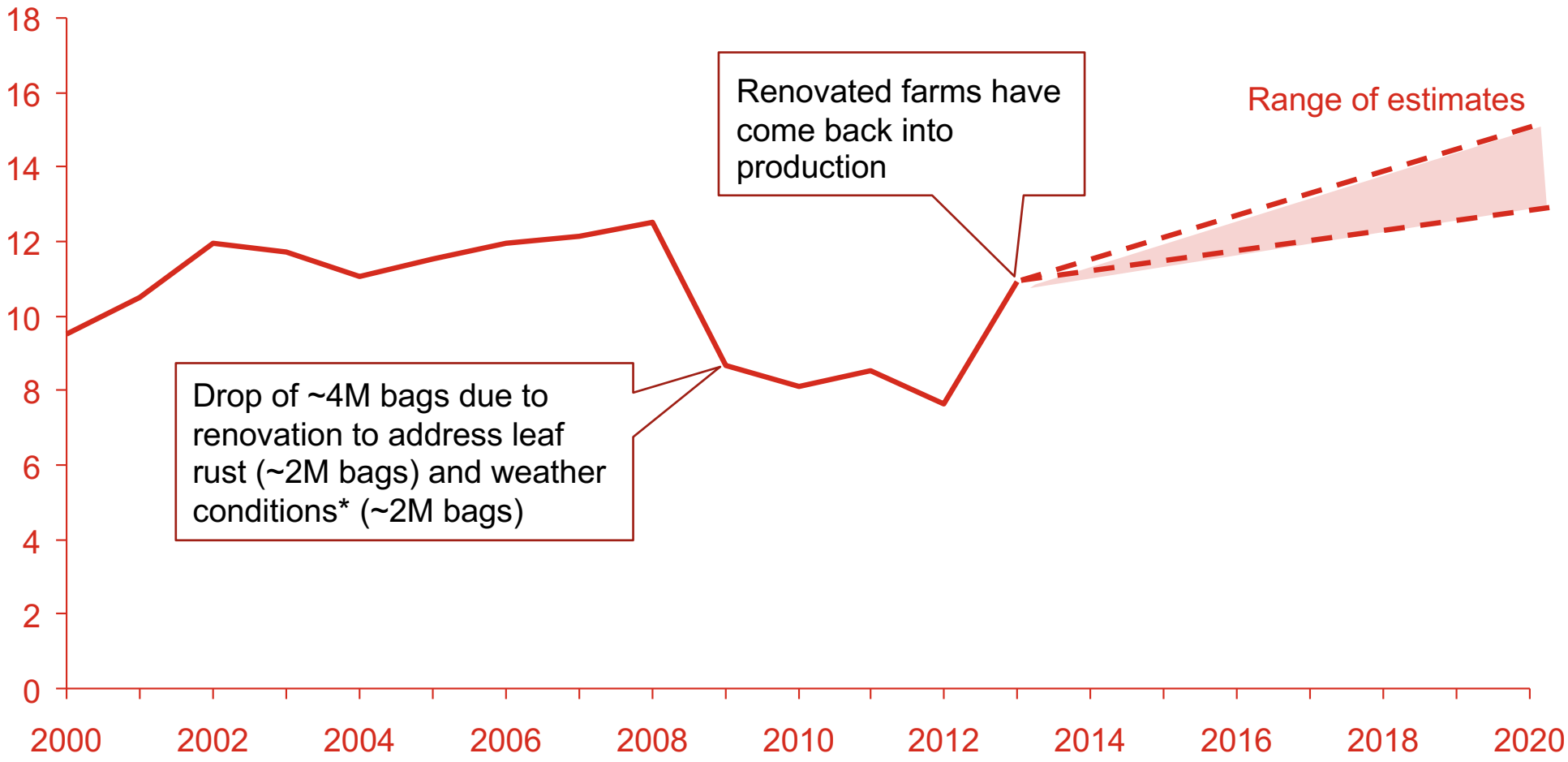
Estimated number of coffee farms
Thousands



* Combination of low international coffee prices from 2001 to 2003 ("coffee crisis") and armed conflict caused farmers to abandon land and/or switch to other crops with a quicker cash cycle, including coca, in particular in lower altitudes
** As of 2008 *** Inactive farmers
Source: FNC; FONAC; 1993-1997 National Coffee Survey; Universidad de los Andes – CEDE: " Abandoning Coffee under the Threat of Violence and the Presence of Illicit Crops. Evidence from Colombia"; TechnoServe estimates

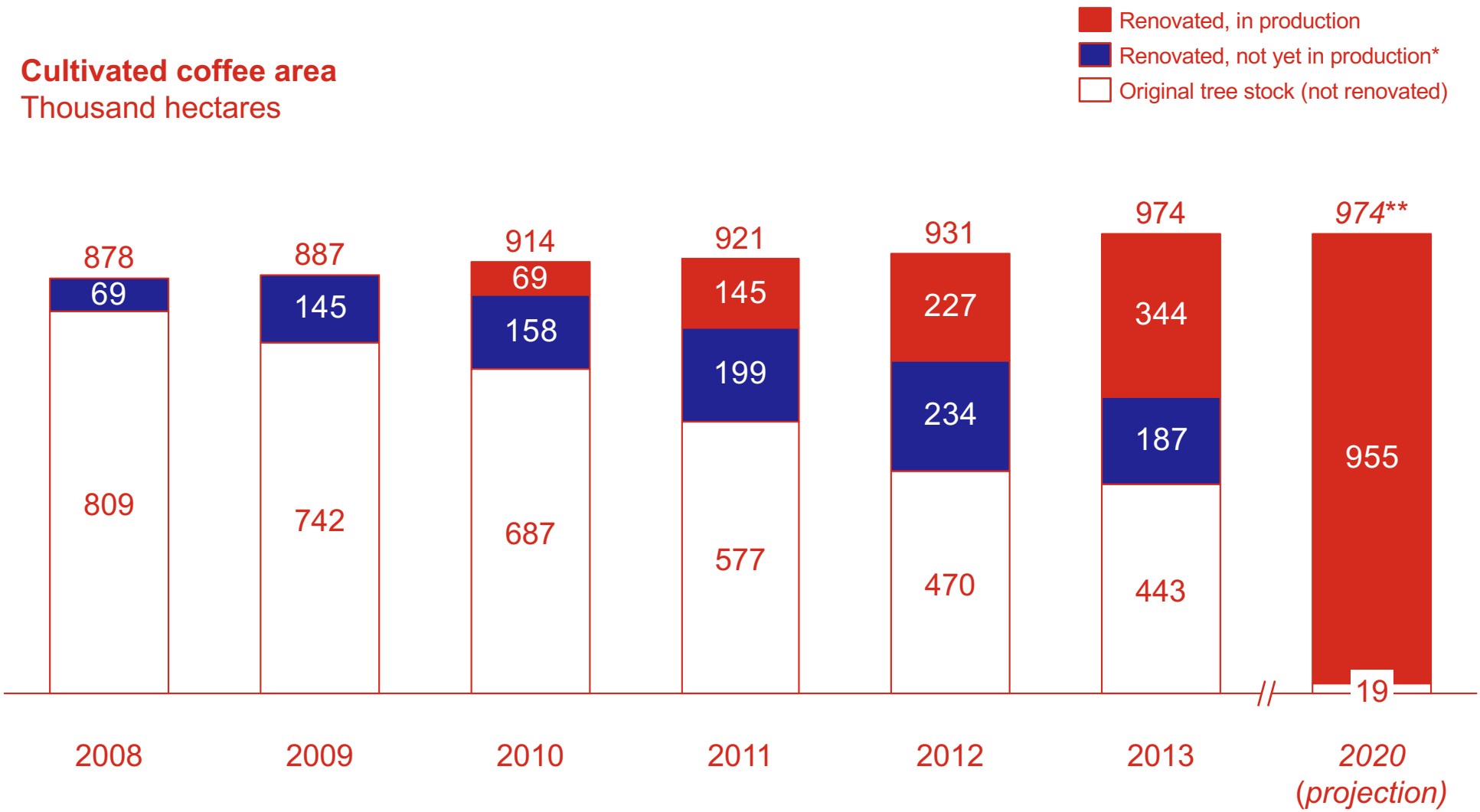
Output fell due to adverse weather and a national tree renovation program, but is now expected to rise

Colombia national coffee production
Millions of bags (60-kg), green



* El Niño and La Niña
Source: USDA FAS; FNC; Interviews

Renovation is ongoing and more than half of existing trees have already been replanted



* Estimate based on assumption that a tree is "in production" in the 3rd harvest after planting
** Assuming that total coffee area remains constant at 2013 value; FNC aims to renovate 98% of tree stock by 2020
Source: FNC

Colombia's supply chain has six distinctive attributes in comparison to other countries

Colombia's coffee supply chain (Arabica)

Distinctive to Colombia

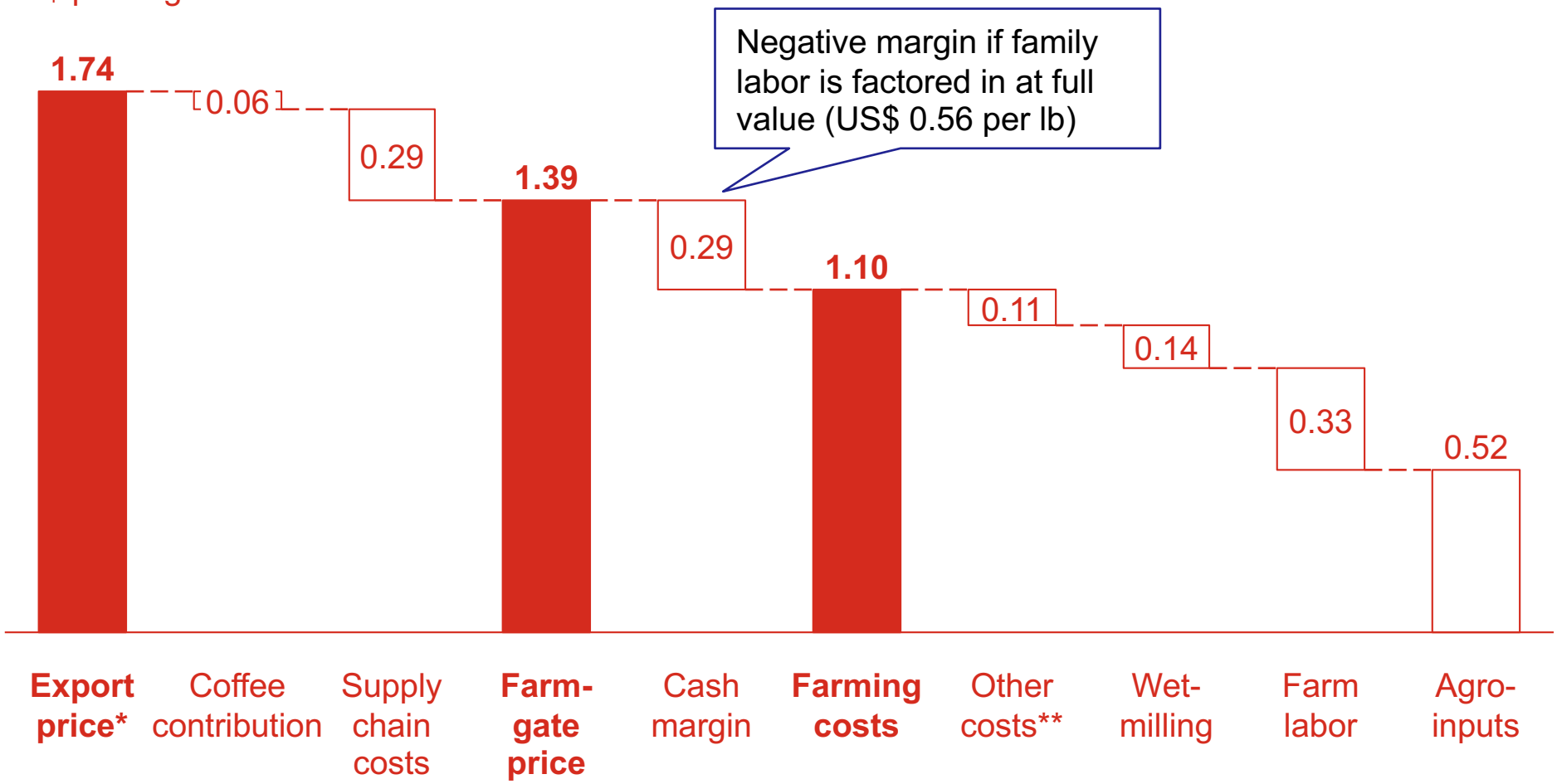
Farming	Processing: Wet milling	Aggregation	Exporting	Roasting
• **Majority of farms are smallholders** (with less than 5 ha under coffee) • **Harvesting occurs year-round** in most areas, with a main harvest and a 2nd mid-year harvest (*mitaca*)	• **Nearly all farmers wet-mill at home and sell dry parchment** (in most regions there is no market for cherry) • Some **wet mills have gaps in infrastructure**, e.g., insufficient drying space, inadequate wastewater disposal, etc.	• Aggregation is done **both by cooperatives and private buyers** • **~ 35% of volume goes through coops*** , the highest share in Latin America • **A purchase guarantee** exists at all aggregation points	• **Competition exists between local and multinational exporters** (multi-national exporters have ~33% market share) • **The FNC**, a nonprofit, is the largest exporter** (25% market share) • **~90% of total production is exported***	• **~10% of production is consumed internally** • **Internal demand is mainly for roast & ground** (rather than soluble) • **Soluble manufacturing capacity exists**, both for export and internal markets

- **Strong institutional framework** through the establishment of the National Coffee Fund, which provides support to the coffee sector and is managed by the FNC

* There are 34 coops with 515 purchase points country-wide; while in 2013 45% of volume went though cooperatives, the share has been ~35% on average over the last years
** Federación Nacional de Café; responsible for the implementation of the National Coffee Fund
*** Coffee not fulfilling export quality standards is used for internal consumption
Source: FNC; interviews

Average farming cash costs are estimated to be around US$1.10 per lb and farmers receive ~ 80% of export price

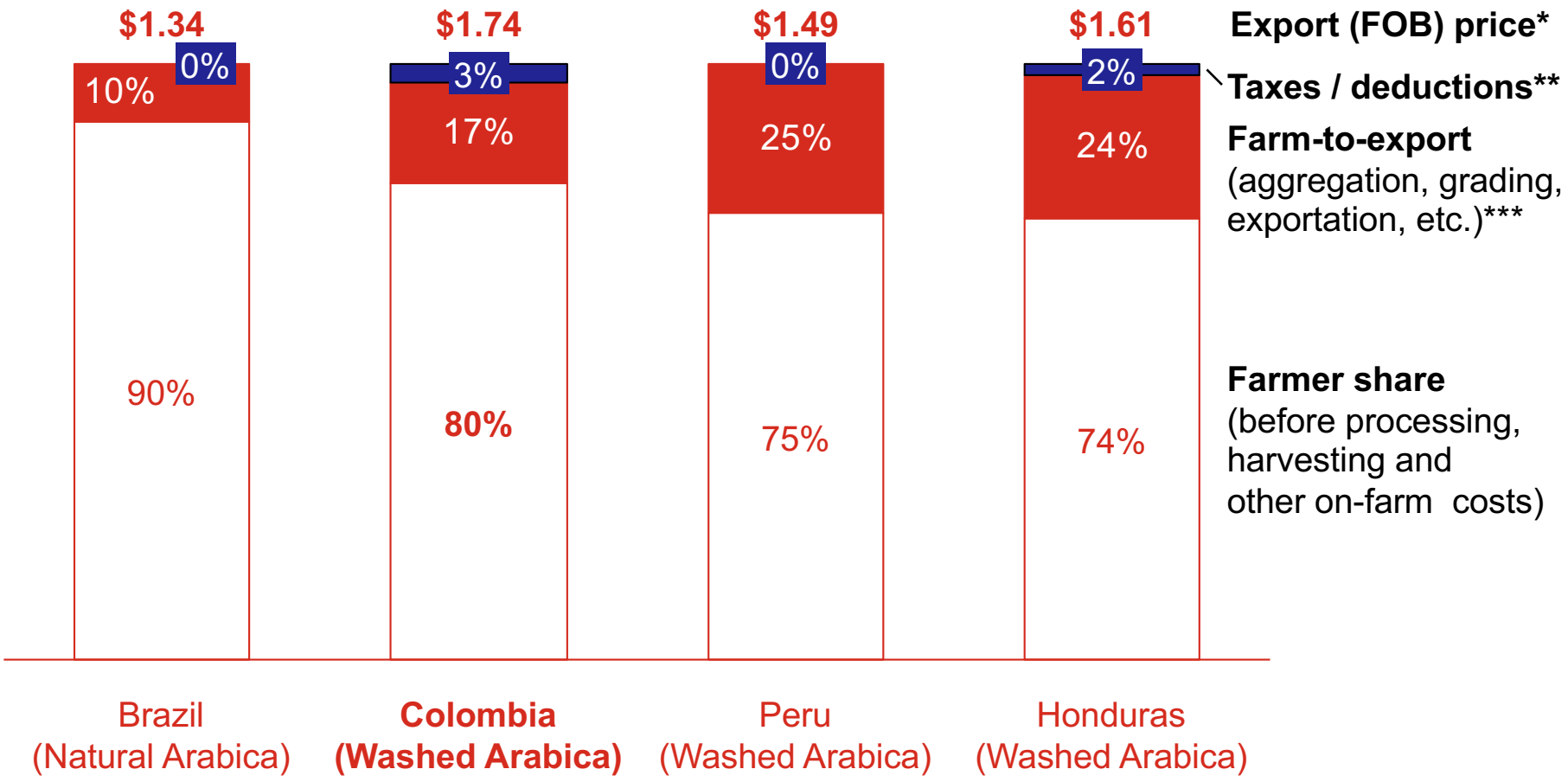
Cash cost of production for average Colombian farm
US$ per lb green



Note: Cost of production does not include costs for installation of wet milling equipment, tree renovation or financing
*Normalized to 2012/13 average commodity prices – Arabica ICE "C" of $1.50 but adjusted for market differentials)
** Includes farm tools, equipment, internal transport, etc.
Source: Stakeholder interviews, TechnoServe analysis

Colombia's farm-to-export chain is efficient relative to regional peers (e.g., Peru, Honduras)

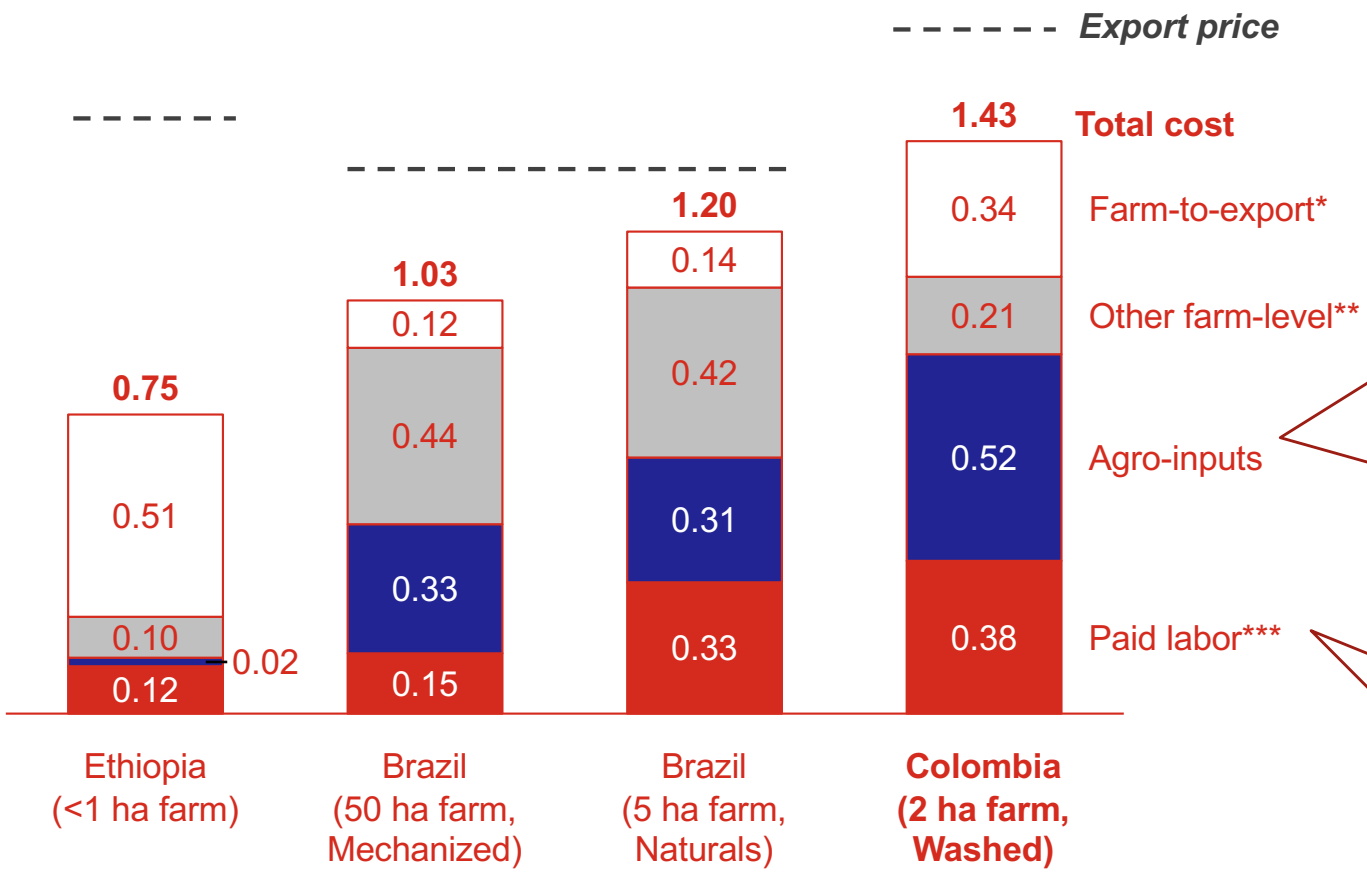
Efficiency levels of major Latin American coffee supply chains, 2012/13
US$ per pound green



*Prices are normalized to 2012/13 average commodity prices – Arabica (ICE "C") $1.50 but adjusted for market differentials
** Coffee contribution of US$ 6 cents/lb in Colombia and US$ 4.25 per 46-kg bag in Honduras (not considering 'retención' of US$ 9 per 46 kg bag that is reimbursed to registered farmers after payment)
*** Calculated as difference between farm gate price and ICO price
Source: trade statistics; TechnoServe analysis and interviews in Brazil, Colombia, Honduras and Peru

However, Colombia's farm-level costs are significantly higher than the two cost leaders (Brazil and Ethiopia)

Cash cost of production breakdown
US$ cent per lb green



Higher agro-input costs (compared to Brazil)
- Less competitive market for agro-inputs
- Prices are 30+% higher than international prices
- No local production
- Lower yields
- Less efficient fertilizer usage

Higher labor costs (compared to Brazil)
- Lower yields
- Lower labor efficiency

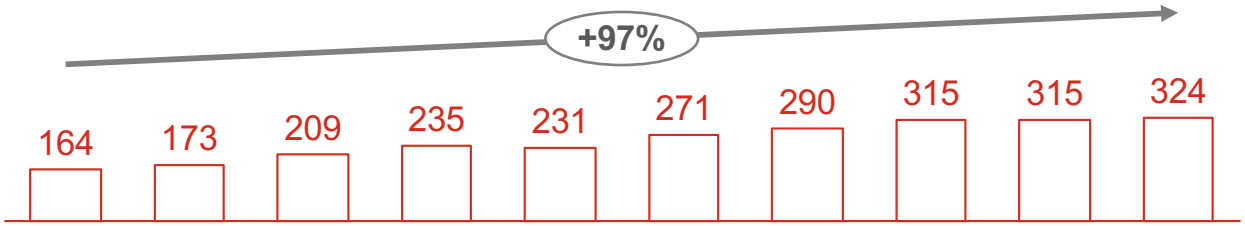
* Includes secondary processing (e.g., dry-milling, grading); also includes primary processing for Ethiopia
* Includes depreciation, fuel, internal transport, other fixed costs
** Excludes family labor
Source: Stakeholder interviews, TechnoServe analysis

Colombia's farming costs have risen over the past decade

x% 2005-14 growth rate

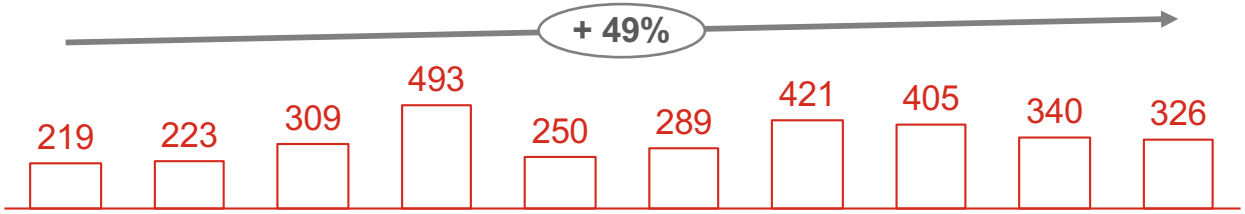
Labor
+97%

Minimum wage (US$ per month)



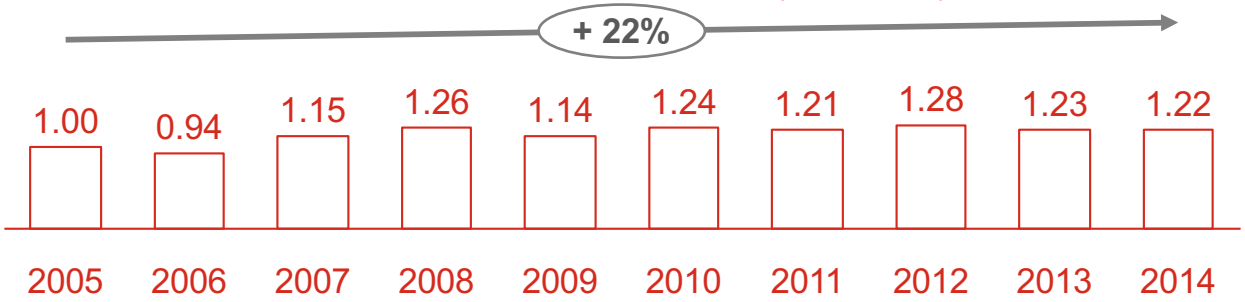
Fertilizer
+49%

Price of urea (US$ per ton)*



Currency
+22%

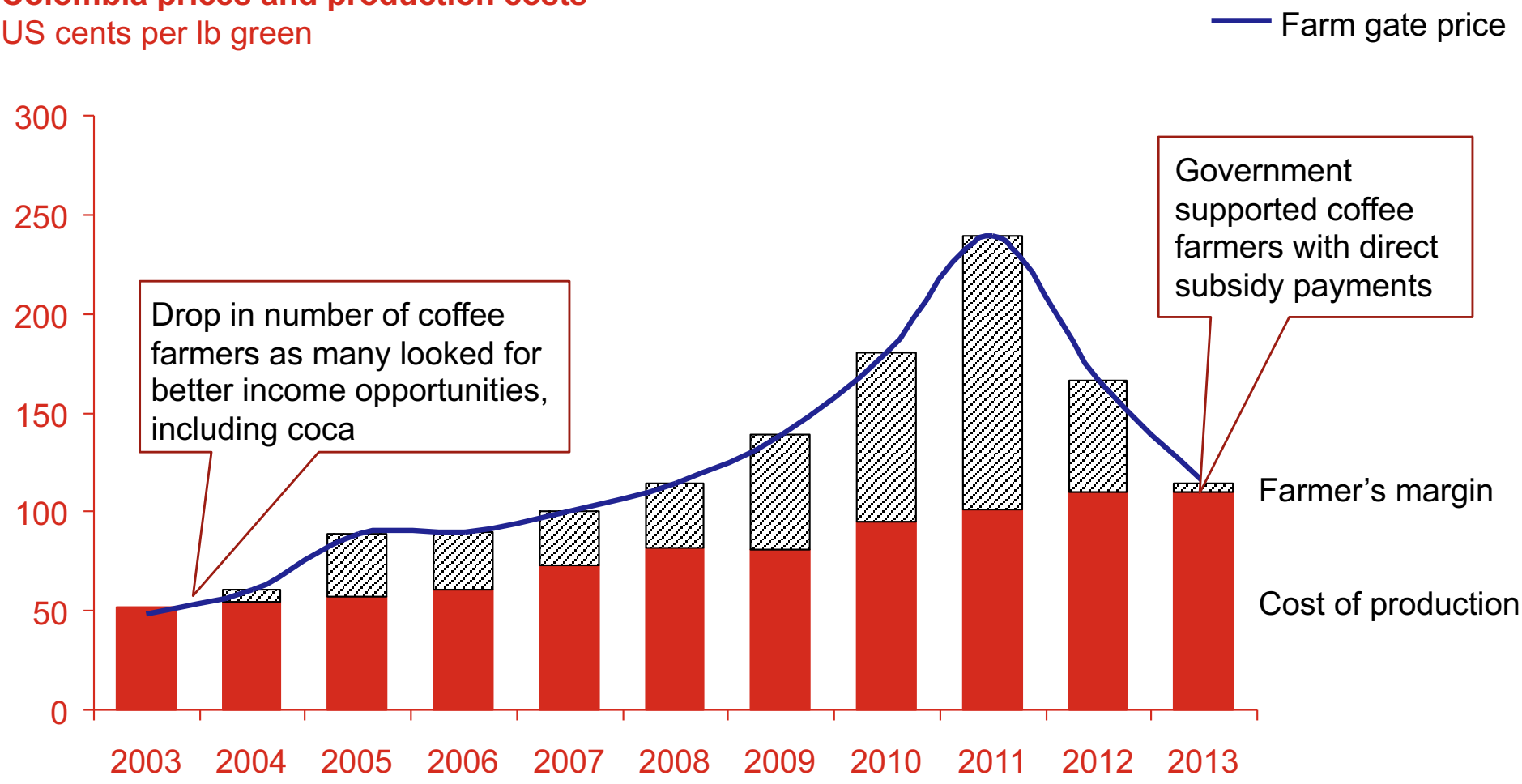
Colombian peso relative to US dollar, indexed (2005=100)**



* Average sales price of key importers; import taxes on fertilizers higher in Colombia than in other coffee producing countries
** Rise is appreciation of the Colombian peso relative to the US dollar
Source: Ministerio de la Protección Social (minimum wage); indexmundi.com (urea price); xe.com (exchange rate)

As prices have fallen, the farmer's margin has been eroded

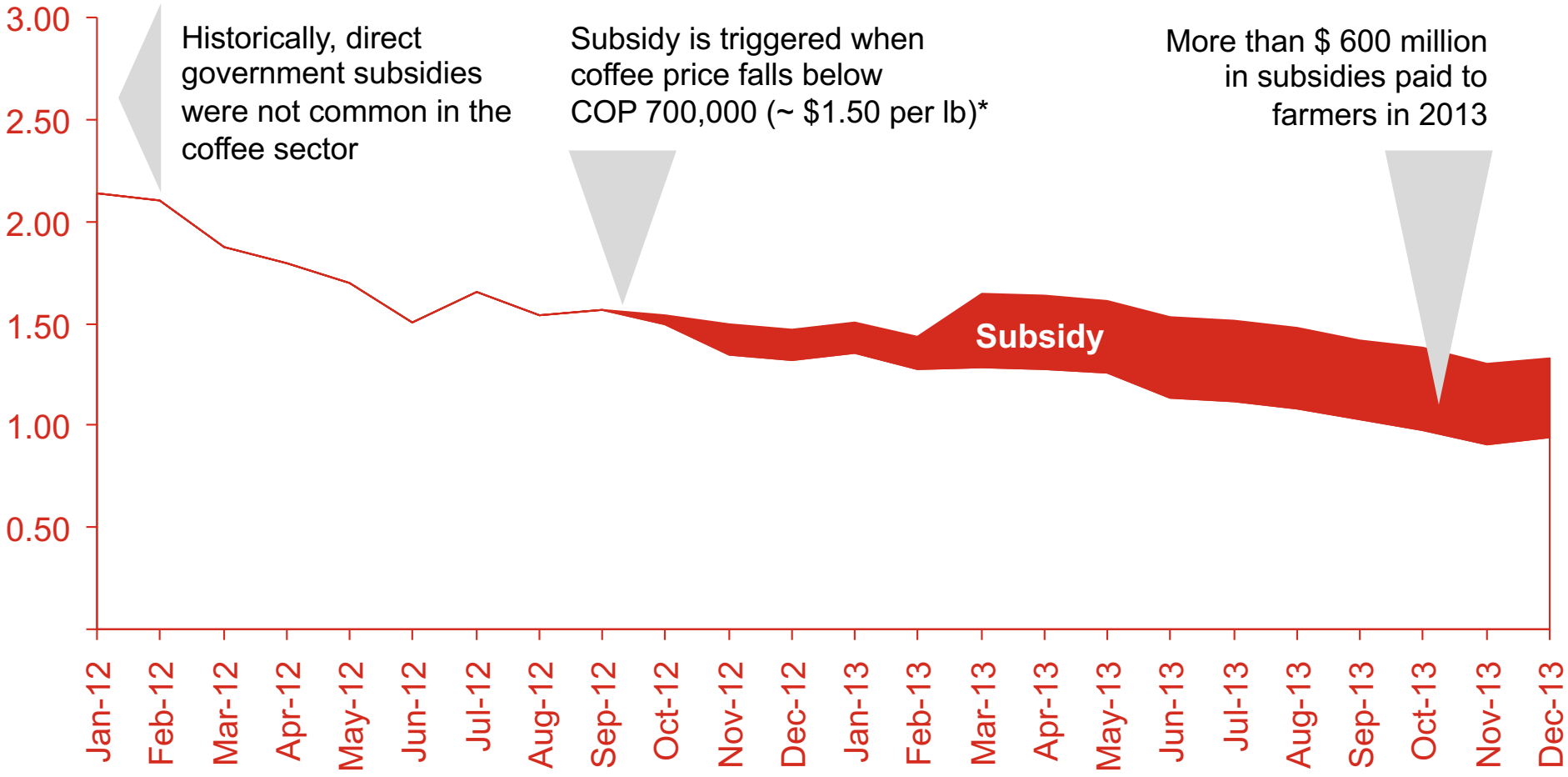
Colombia prices and production costs*
US cents per lb green



* Cash production costs
Source: ICO, FNC, TechnoServe analysis

The government has helped farmers by offering a cash subsidy when the market falls below US$1.50 (COP 700,000)

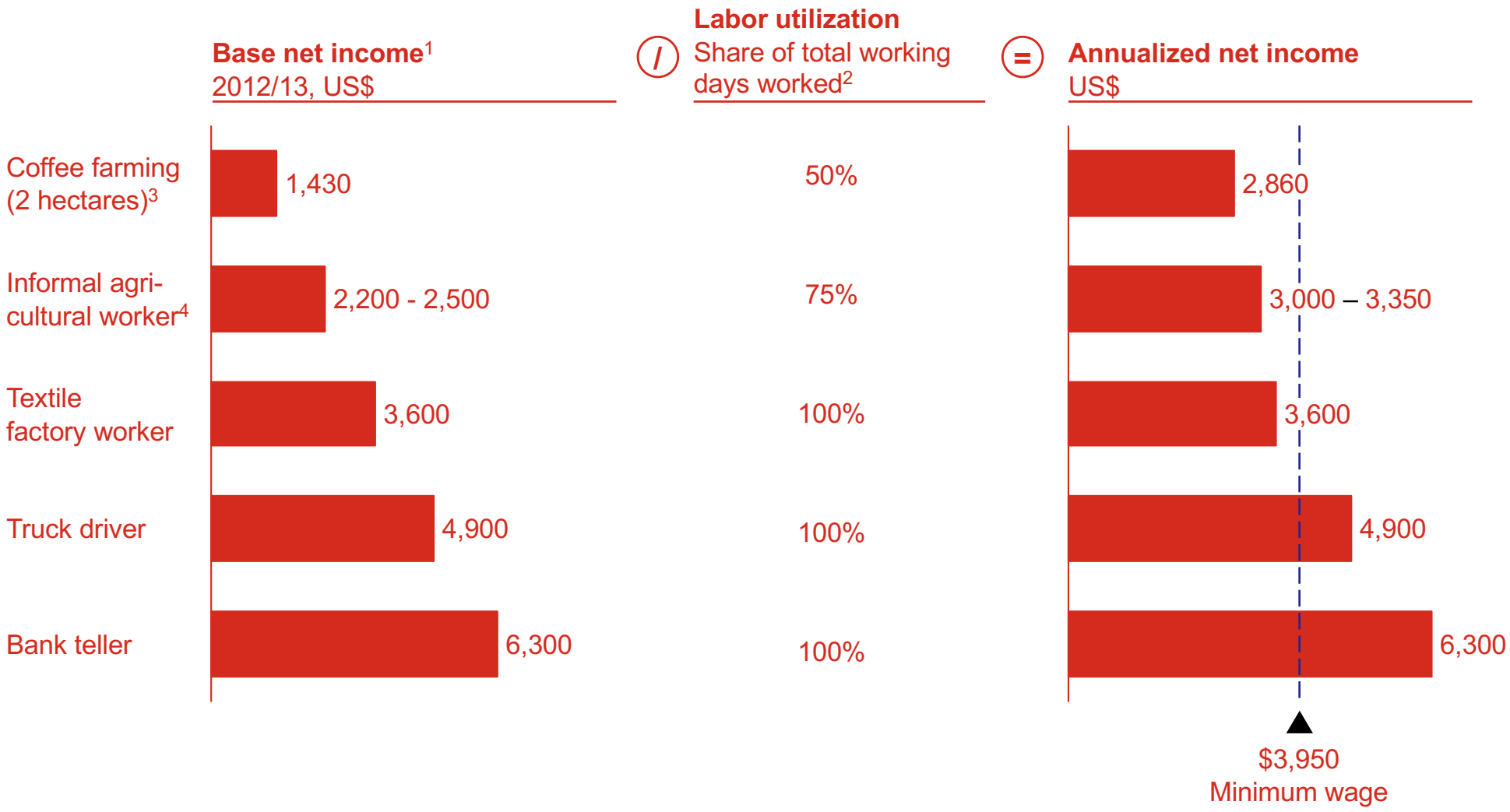
Colombia average farm-gate price
US$ per lb green



*At current exchange rate; subsidy paid if internal base price is below COP 700,000; sum of income from coffee sale and subsidy cannot exceed COP 700,000; subsidy of COP 165,000 paid if price is below COP 480,000; otherwise payment of up to COP 145,000; between Oct 2012 and March 2013 producers received COP 20,000 and later 60,000 per load if price was below 650,000

Source: FNC, TechnoServe analysis

Farmers will increasingly weigh the opportunity cost of coffee farming, both in terms of land and time (labor)



1 Entry level salary
2 Assuming 260 work days per year for office job and 300 days for agricultural workers
4 Assuming daily rate of COP 20,000 and 220 work days per year
Source: DANE, tusalario.org, stakeholder interviews, TechnoServe analysis

Colombia is at a crossroads

Facing eroding margins and decreasing competitiveness, Colombian coffee farmers have several options to address this situation

Slowly exit coffee

- Shift away from coffee farming and move into other economic activities that provide better income prospects
- In particular, in peri-urban areas and in places most vulnerable to climate change

Keep coffee but diversify

- Diversification into additional crops that provide income and decrease dependency on coffee prices
- In particular, in lower altitudes that are more susceptible to disease and climate change

Focus on quality

- Focus on specialty coffee and implement activities to improve quality in order to achieve higher margins
- In particular for farmers at higher altitudes or with traditional varieties

Improve efficiency

- Decrease cost of production in order to improve efficiency and gain higher profit margin

Source: Stakeholder interviews, TechnoServe analysis

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- ❑ **Analytical framework**
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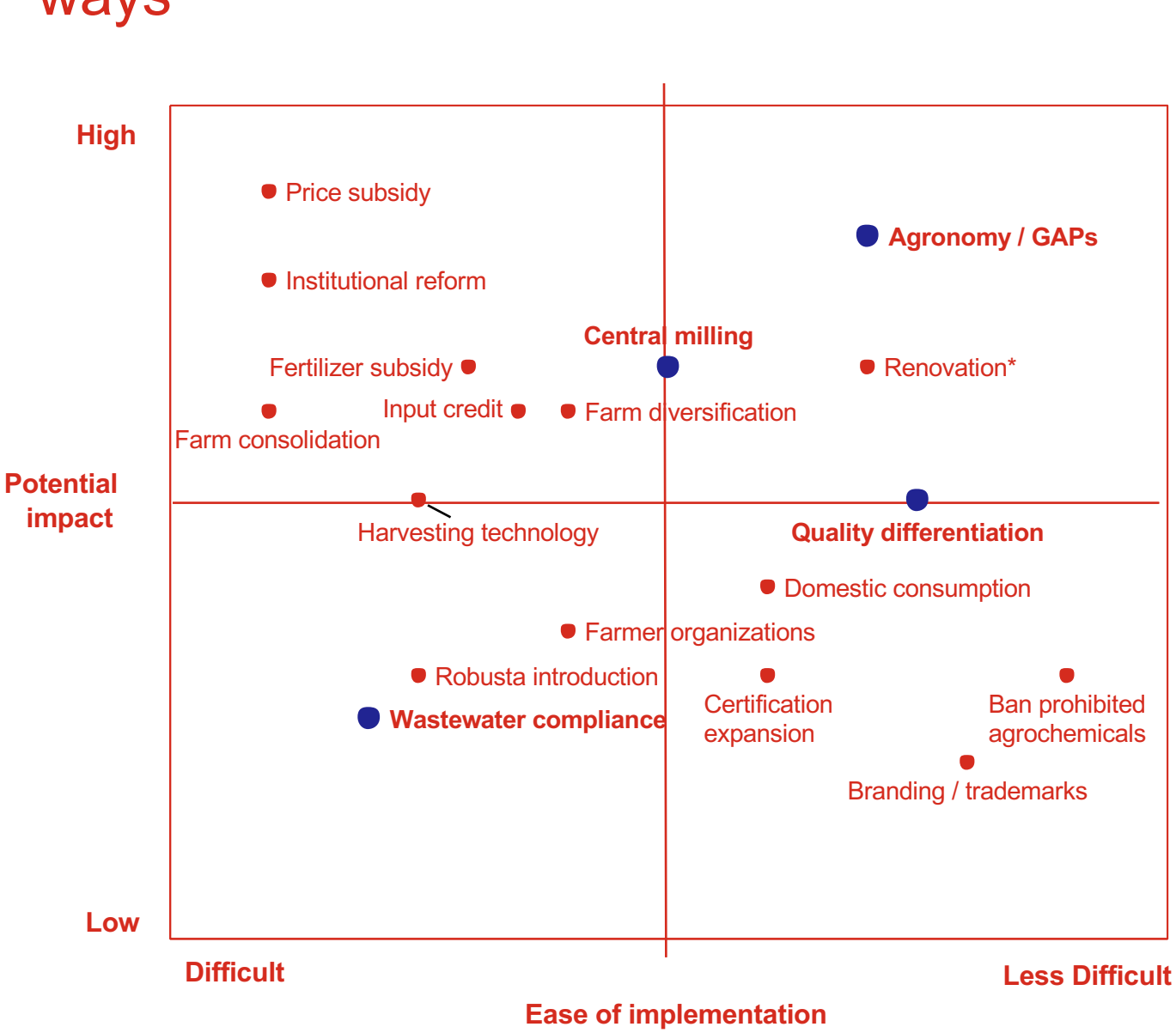
Different groups of farmers will need to respond in different ways



Potential Impact
- Net Benefit / ROI
- Revenue – GDP increase
- Employment – Potential for job creation or preservation
- Scope – Broadly applicable or impacting a narrow segment
- Durability– Ability to preserve jobs/wealth over 5-20 year time frame

Ease of Implementation
- Time Frame – Time required to generate economic growth
- Cost – Investment required
- Political Feasibility – Ability to reach stakeholder agreement and long term commitment
- Technical Feasibility – Complexity of solution and availability of local skills to implement

Interpreting this matrix
- "High Impact / Less Difficult" items are recommended co-investments
- "High Impact / Difficult" items are worth pursuing but likely require long-term funding or should be government led
- "Low Impact / Less Difficult" items are likely to happen on their own (i.e., via private sector or existing initiatives)
- "Low Impact / Difficult" items should be considered on a case-by-case basis but may not be replicable to all farmers

* Addressed by different schemes
Source: Stakeholder interviews, TechnoServe analysis

We have segmented Colombia's farms on the basis of region, farm size and altitude

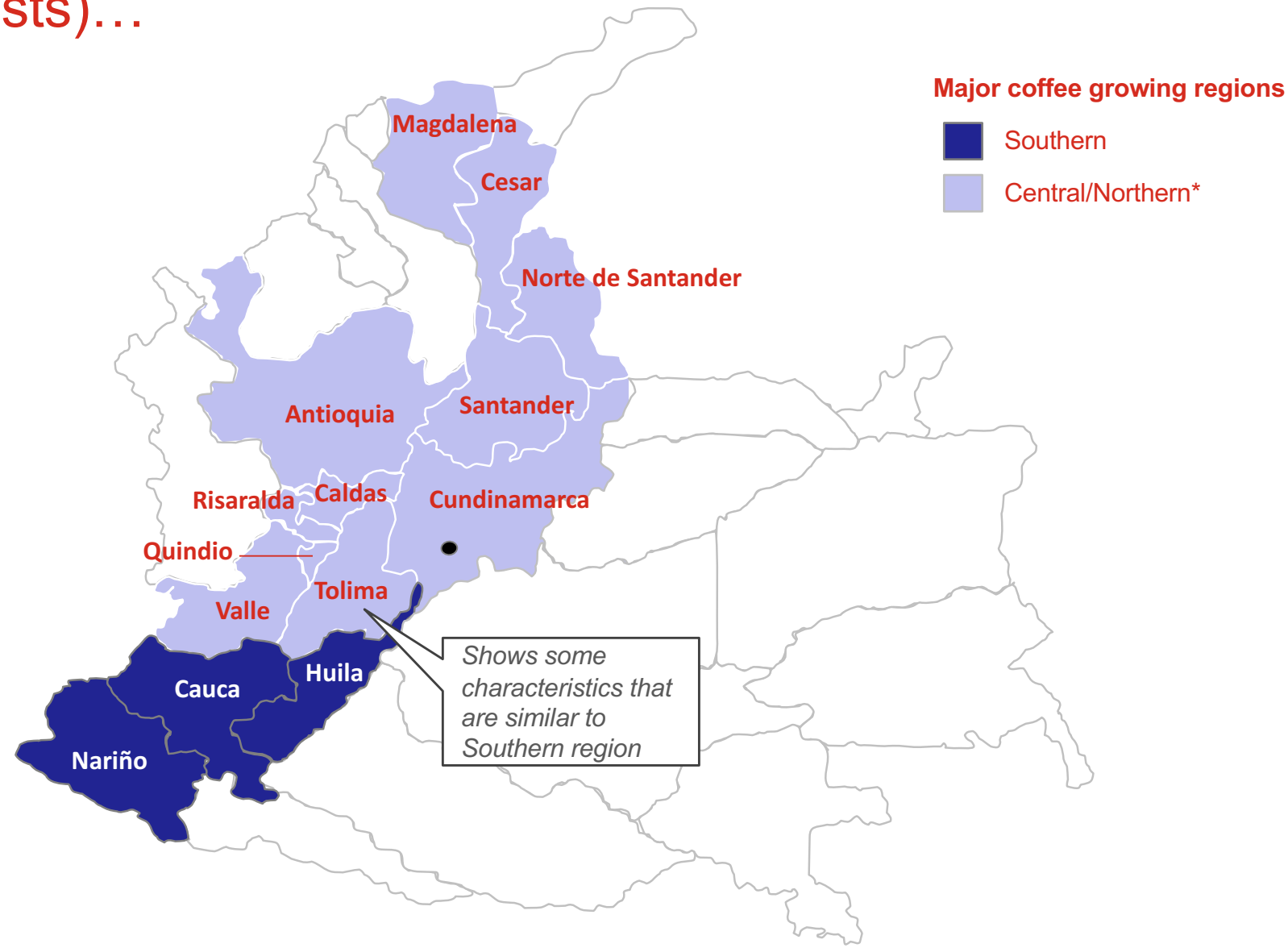
	Implication	Categories
A Region	Some regions have higher labor and opportunity costs (availability of other income opportunities), as well as better infrastructure (e.g., roads)	• Central/northern region • Southern region
B Farm size	Smaller farms typically have lower labor costs because they primarily utilize (unpaid) family labor	• < 1ha • 1 – 5 ha • > 5 ha
C Altitude	Higher altitude areas are more likely to produce premium coffees; lower altitude areas are more vulnerable to pest/disease and climate change	• Low (< 1,200m) • Medium (1,200 – 1,600m) • High (> 1,600m)

Source: Stakeholder interviews, TechnoServe analysis

A

REGIONAL ARCHETYPES

Southern farms tend to have lower opportunity costs (and labor costs)…

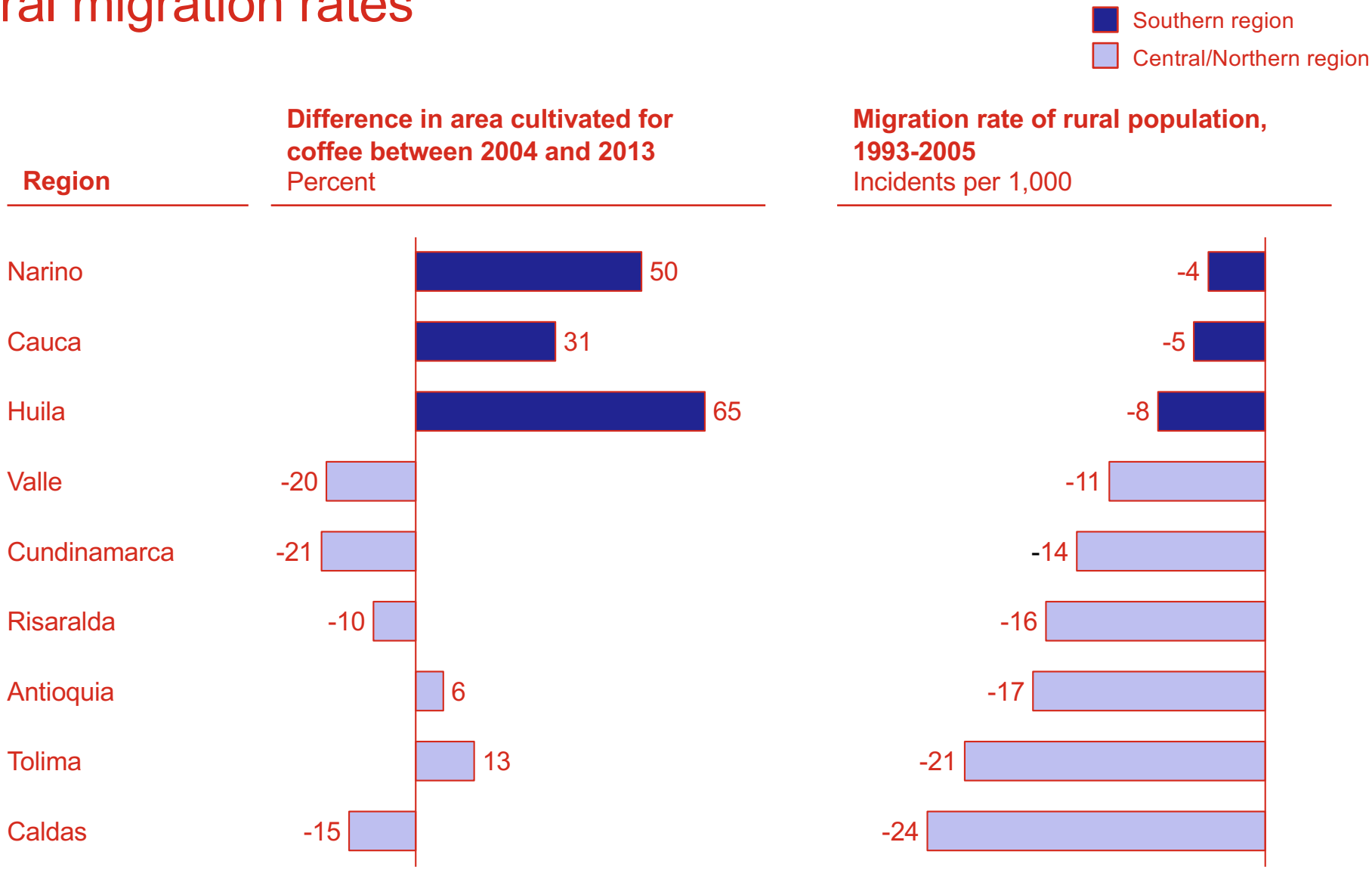


Note: Regional classification based on Cenicafé definition
*Inclusion of northern regions into central region despite regional differences
Source: Agronet, Cenicafé

A

REGIONAL ARCHETYPES

…Evidenced in part by growing coffee areas and lower rural migration rates

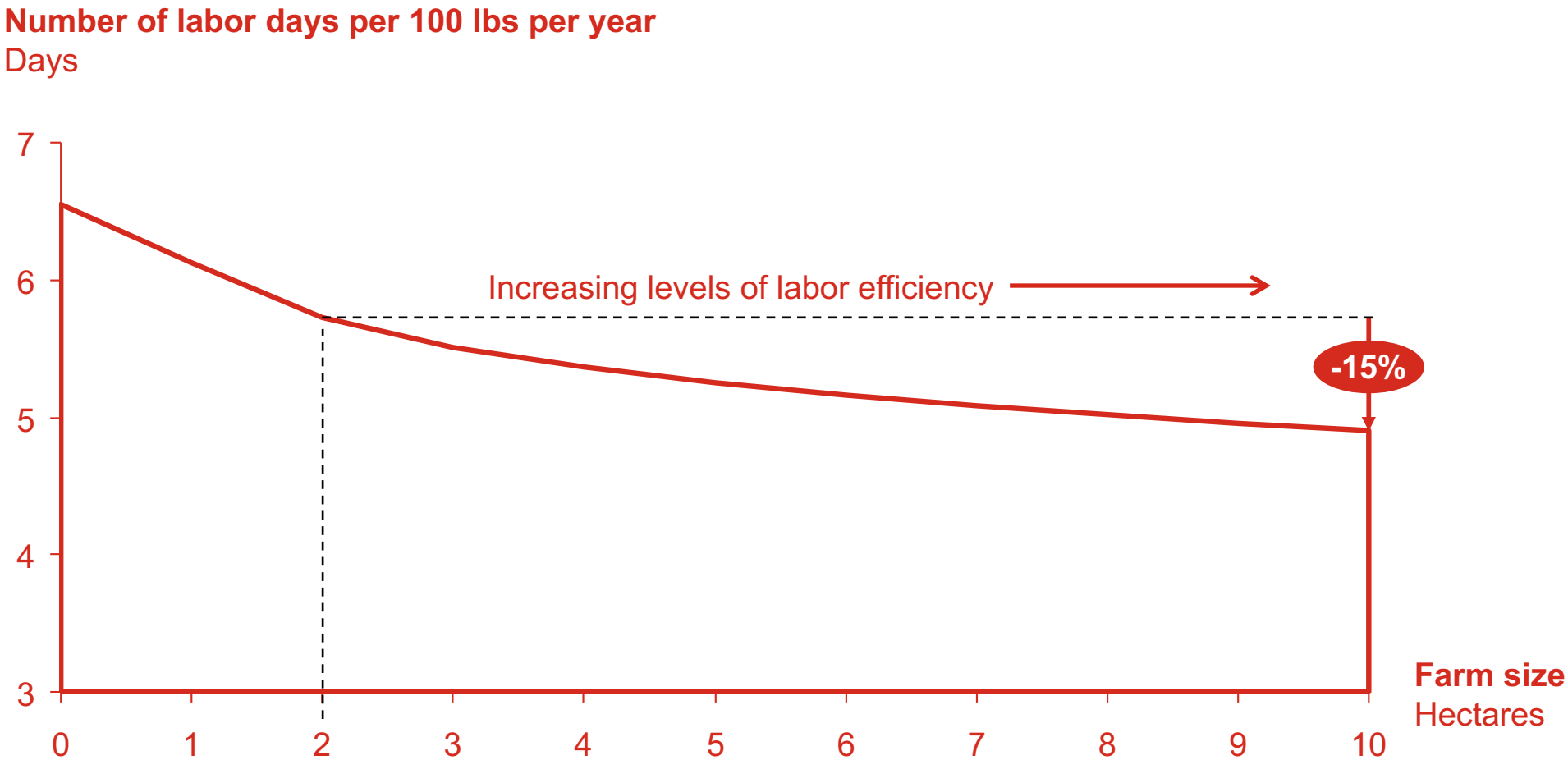


Source: FNC, DANE, TechnoServe analysis

B

FARM SIZE ARCHETYPES

Larger farms tend to utilize labor more efficiently



Source: Interviews; TechnoServe analysis

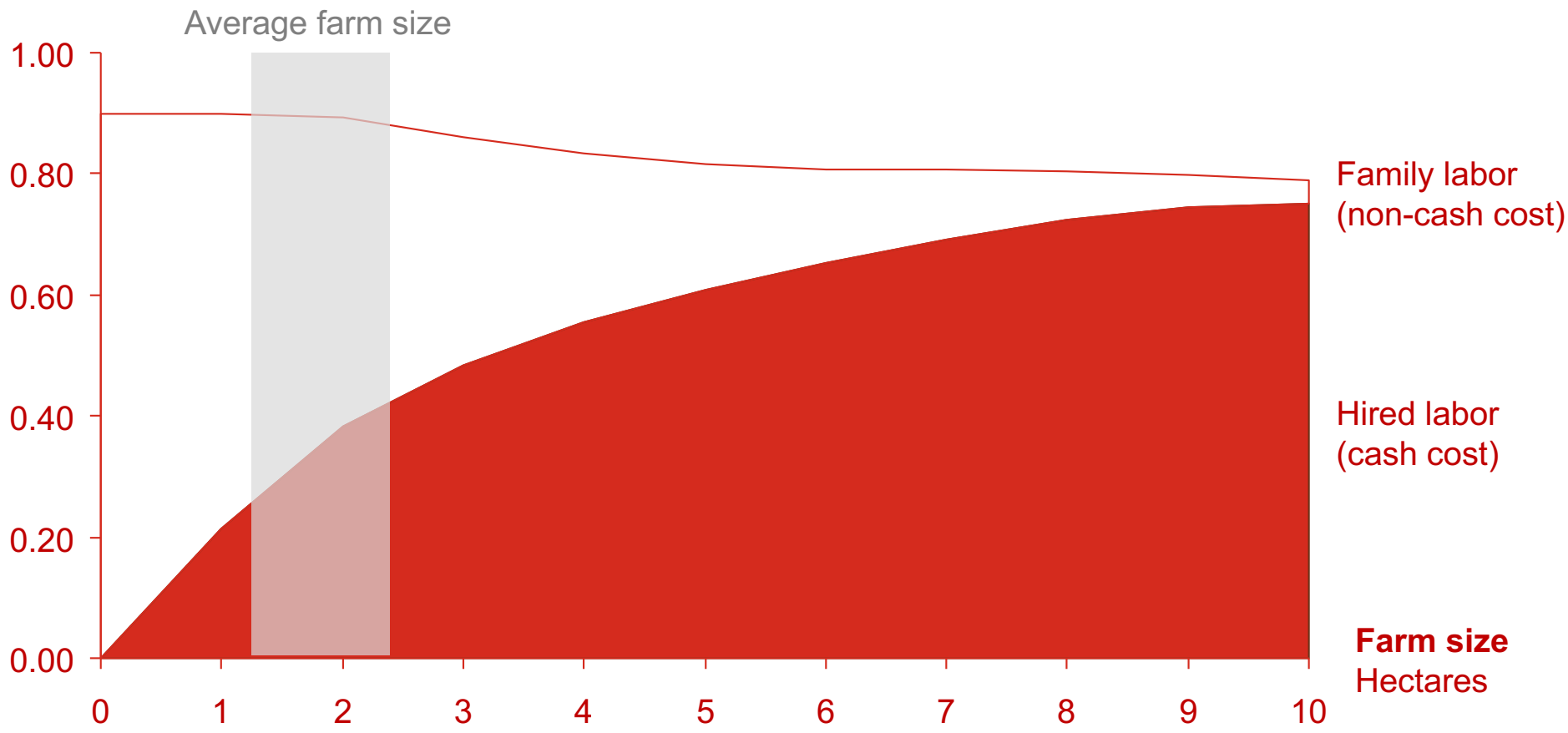
B

FARM SIZE ARCHETYPES

Smaller farms, which rely mostly on family labor, tend to have lower cash costs

Estimated distribution of family vs. hired labor by farm size

US$ per lb green



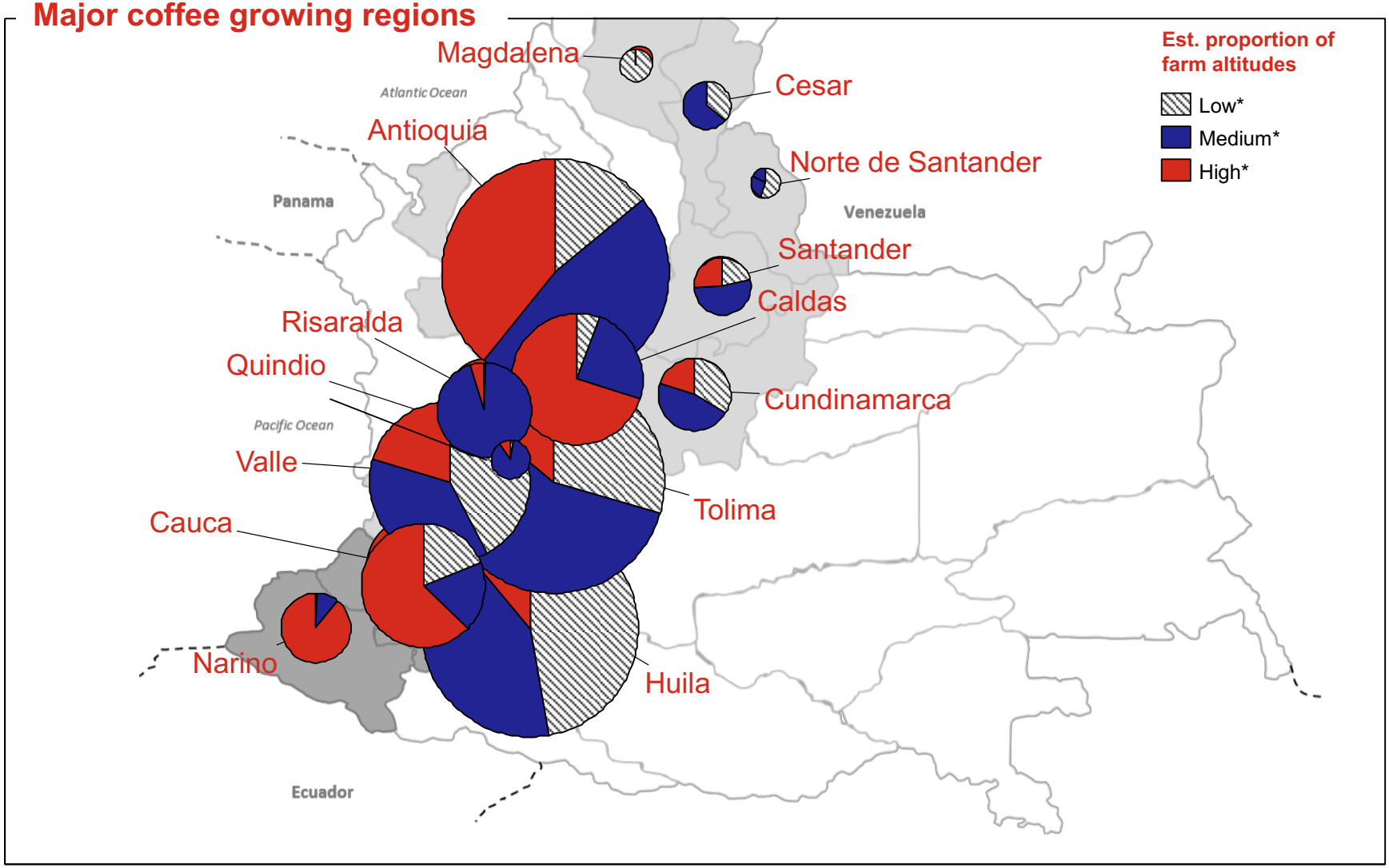
Source: Interviews; TechnoServe analysis

C

ALTITUDINAL ARCHETYPES

Most of Colombia's major growing regions exhibit a range of altitudes

Major coffee growing regions



Note: Usage of average altitude range of 1,600m for specialty coffee; does not account for fact that altitudes might have different meanings in different departments

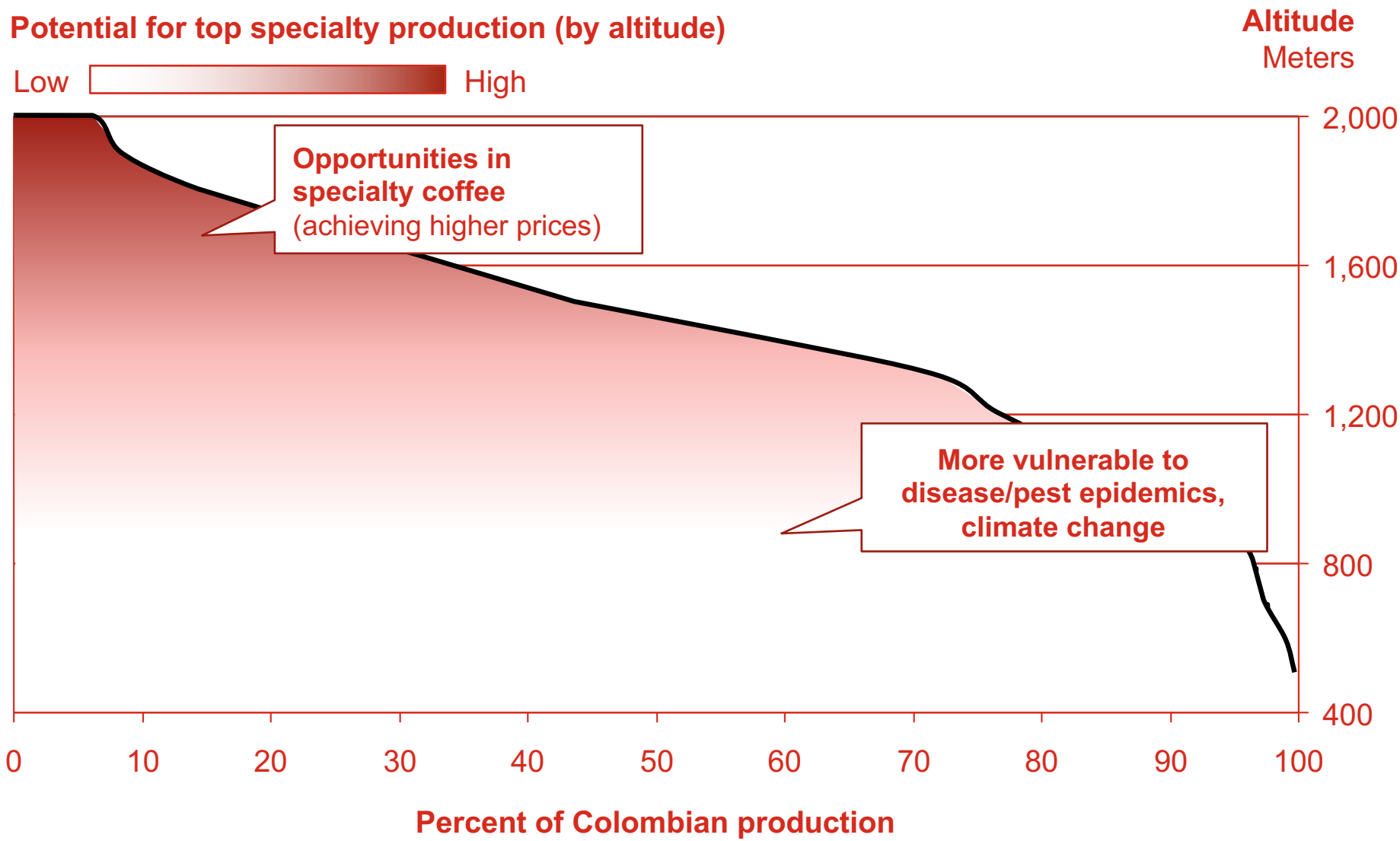
* Low: < 1,200m; medium: 1,200 – 1,600m; high: > 1,600m

Source: AGRONET, FNC, TechnoServe analysis

C

ALTITUDINAL ARCHETYPES

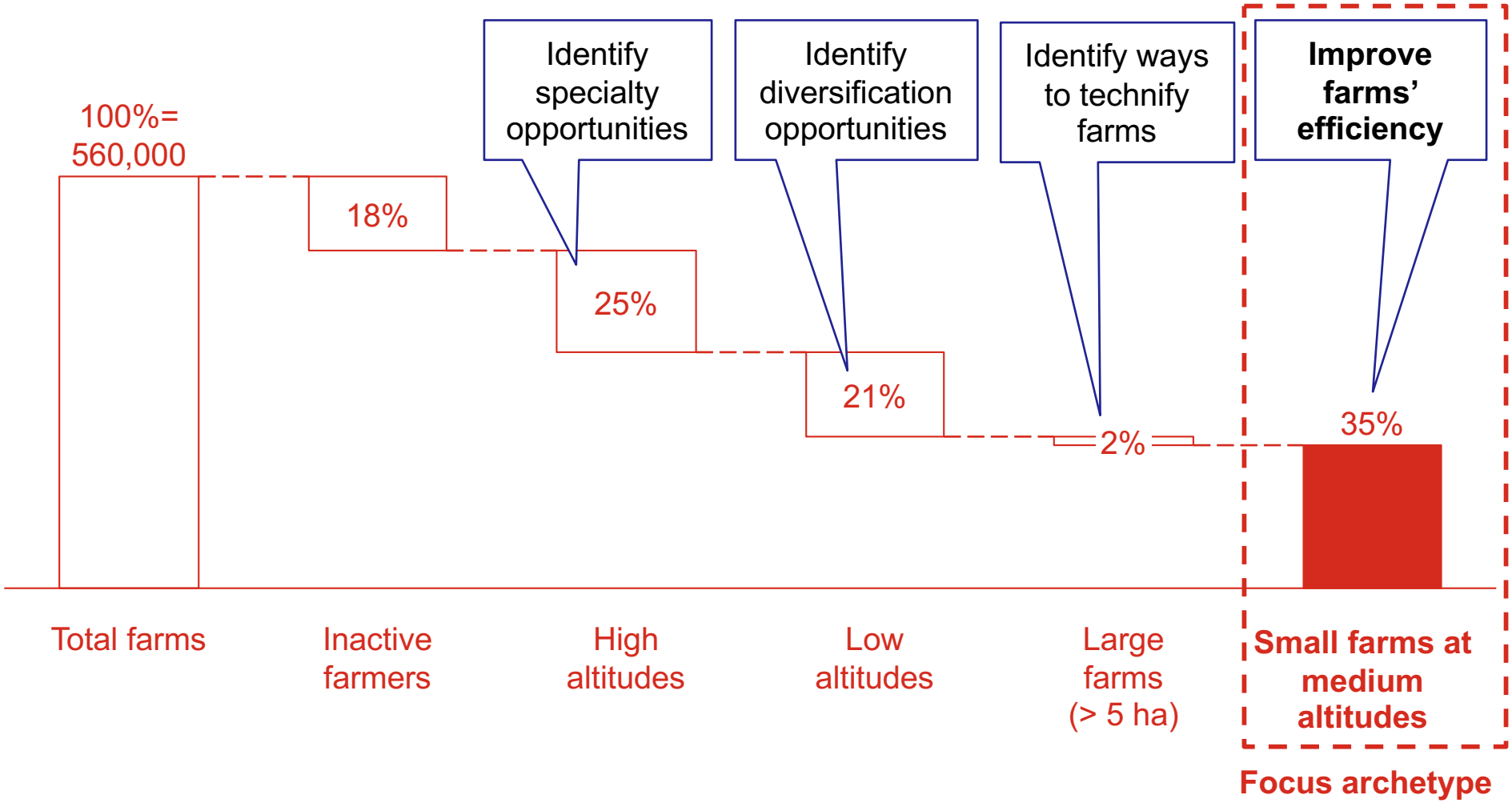
Higher altitude farms have opportunities to focus on top specialty production while lower altitude farms are more vulnerable



Source: AGRONET, FNC, TechnoServe analysis

PRIORITIZING ARCHETYPES

Our solution approach will focus on active farmers in the “mainstream” archetype



Source: TechnoServe analysis

ESTABLISHING THE BUSINESS CASE STUDIES

Three business cases were analyzed in depth targeted at mainstream archetype

Business cases and key challenges

1 Farmers can increase yields through improved agronomy and technical assistance

- Colombia's farm yields could be increased by 20 to 30% by transferring improved agronomy practices to farmers, in particular effective fertilizer usage
- Achieving yield improvement requires high adoption level of best practices; high average age of farmer population could pose a challenge
- Higher yields contribute to a 5% cost reduction and net income increase of 30-50%
- Overall, production would increase by 1.5M bags and exports by US$ 260M per year
- Cost of US$ 200M to train 50% of active farmers (230,000) over 10 years
- Training should complement FNC's individual farm extension visit s with a "farmer field schools" methodology and should actively try to involve the youth

2 Production costs could be reduced modestly through various sector efficiencies

- Colombia's coffee farming is already quite sophisticated and shows limited potential for reducing production costs
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- For harvesting, there are conceptually opportunities for improving productivity; however, implementation will be difficult and results may not justify costs
- For wet-milling, the introduction of central milling could reduce labor requirements by 50-70% and produce other financial and nonfinancial benefits
- Farmers facing the highest opportunity costs to time and labor should be prioritized

3 Supply chains can meet stricter environmental standards with minimal added cost

- Challenging for farms to meet new, stricter environmental regulations – requires new wastewater treatment systems that are complicated and expensive
- Ensure compliance with environmental laws cost-effectively through central wet mills; a single system now benefits many farms

- *The combination of all three business cases would increase net income by more than 50% and reduce time spent for coffee farmers, enabling them to earn more from coffee than other rural employment options*

Contents

- ❑ Colombia coffee industry primer
- ❑ Analytical framework
- ❑ **Colombia business case discussion**

BUSINESS CASE DISCUSSION

Business case 1: Improving yields

Business cases and key challenges

1 Farmers can increase yields through improved agronomy and technical assistance

- Colombia's farm yields could be increased by 20 to 30% by transferring improved agronomy practices to farmers, in particular effective fertilizer usage
- Achieving yield improvement requires high adoption level of best practices; high average age of farmer population could pose a challenge
- Higher yields contribute to a 5% cost reduction and net income increase of 30-50%
- Overall, production would increase by 1.5M bags and exports by US$ 260M per year
- Cost of US$ 200M to train 50% of active farmers (230,000) over 10 years
- Training should complement FNC's individual farm extension visit s with a "farmer field schools" methodology and should actively try to involve the youth

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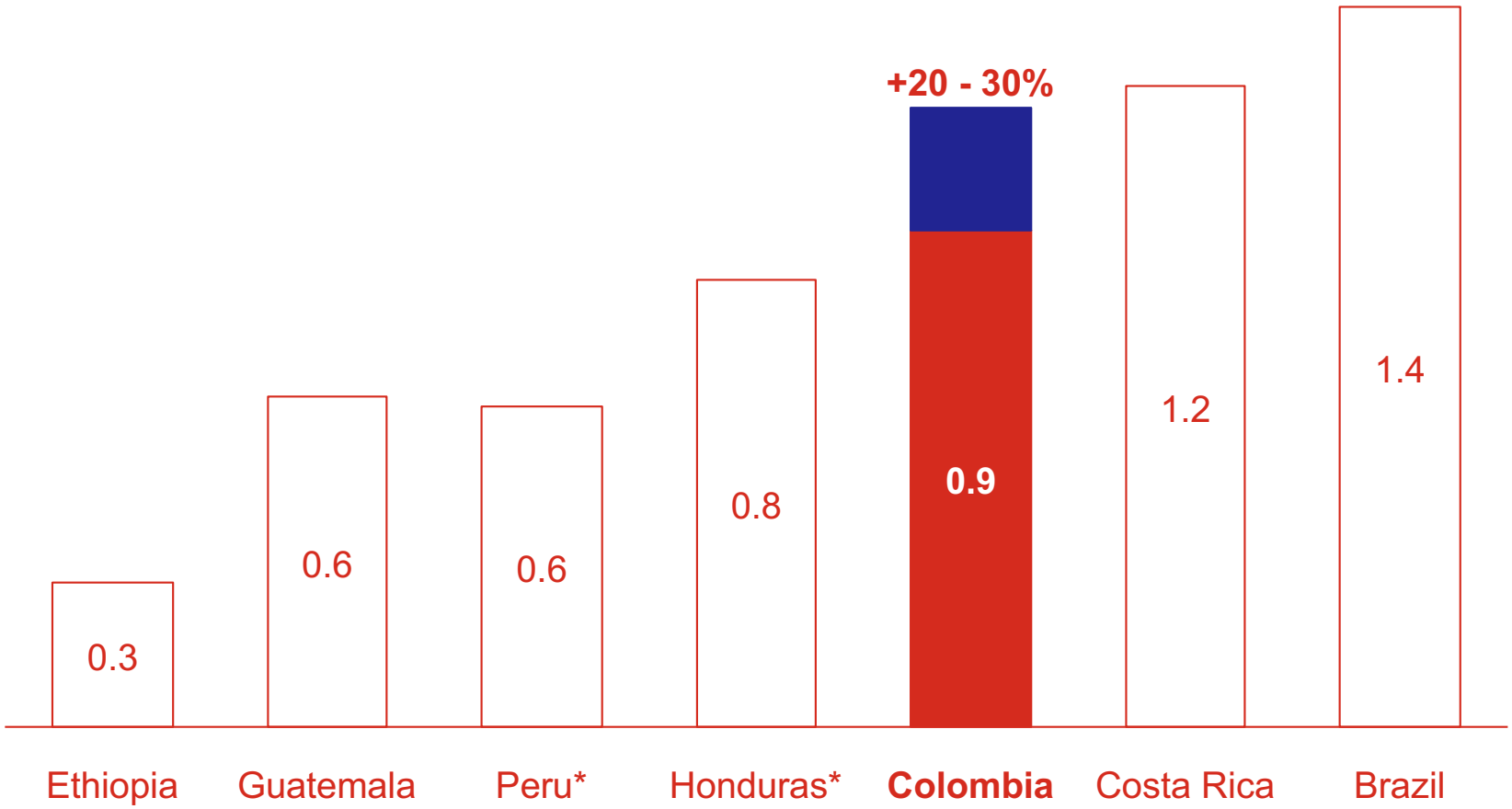
1 Colombia's farm yields could be increased by 20 to 30%

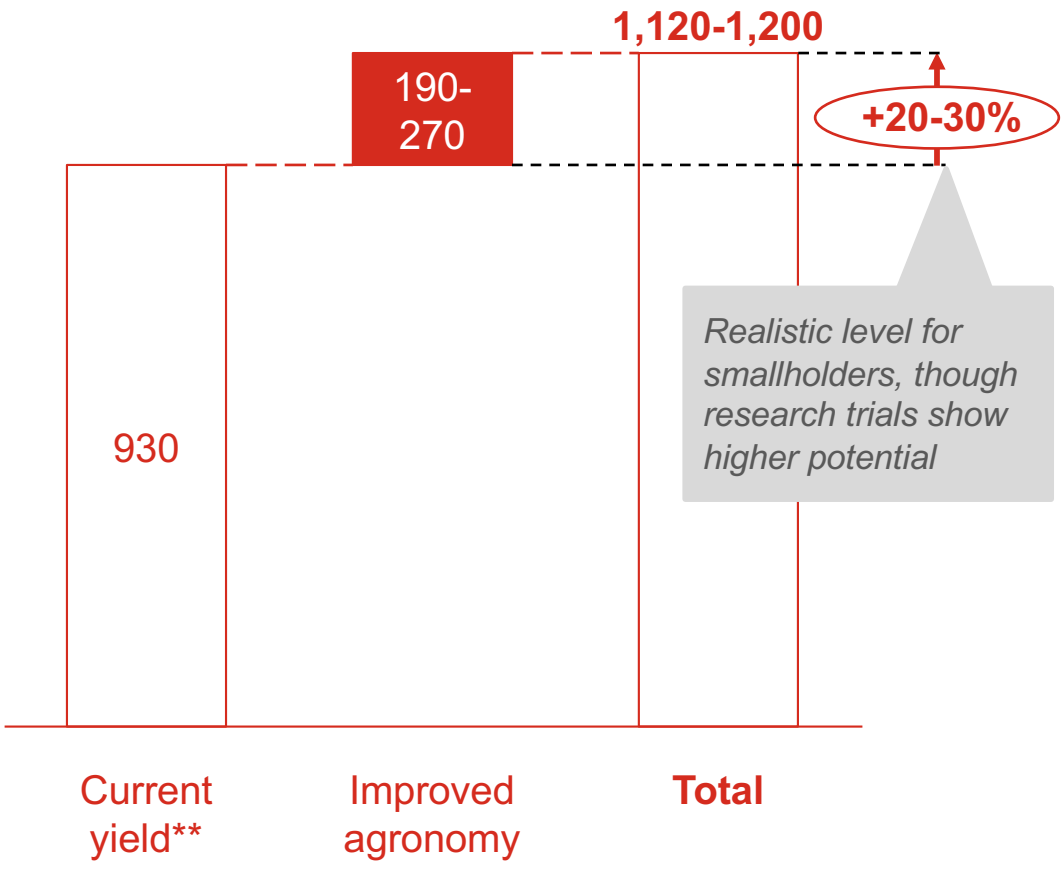
Average coffee farm yields (Arabica)
Metric tons green coffee per ha



* Drop of yield in recent years due to leaf rust
Source: ICO (production volume), national coffee associations and stakeholder interviews for coffee area information

1 Improved agronomy can boost yields by 20 to 30%

Yields could increase by 20 to 30%*…
Kg green coffee per ha



… Through improved adoption of agronomic best practices

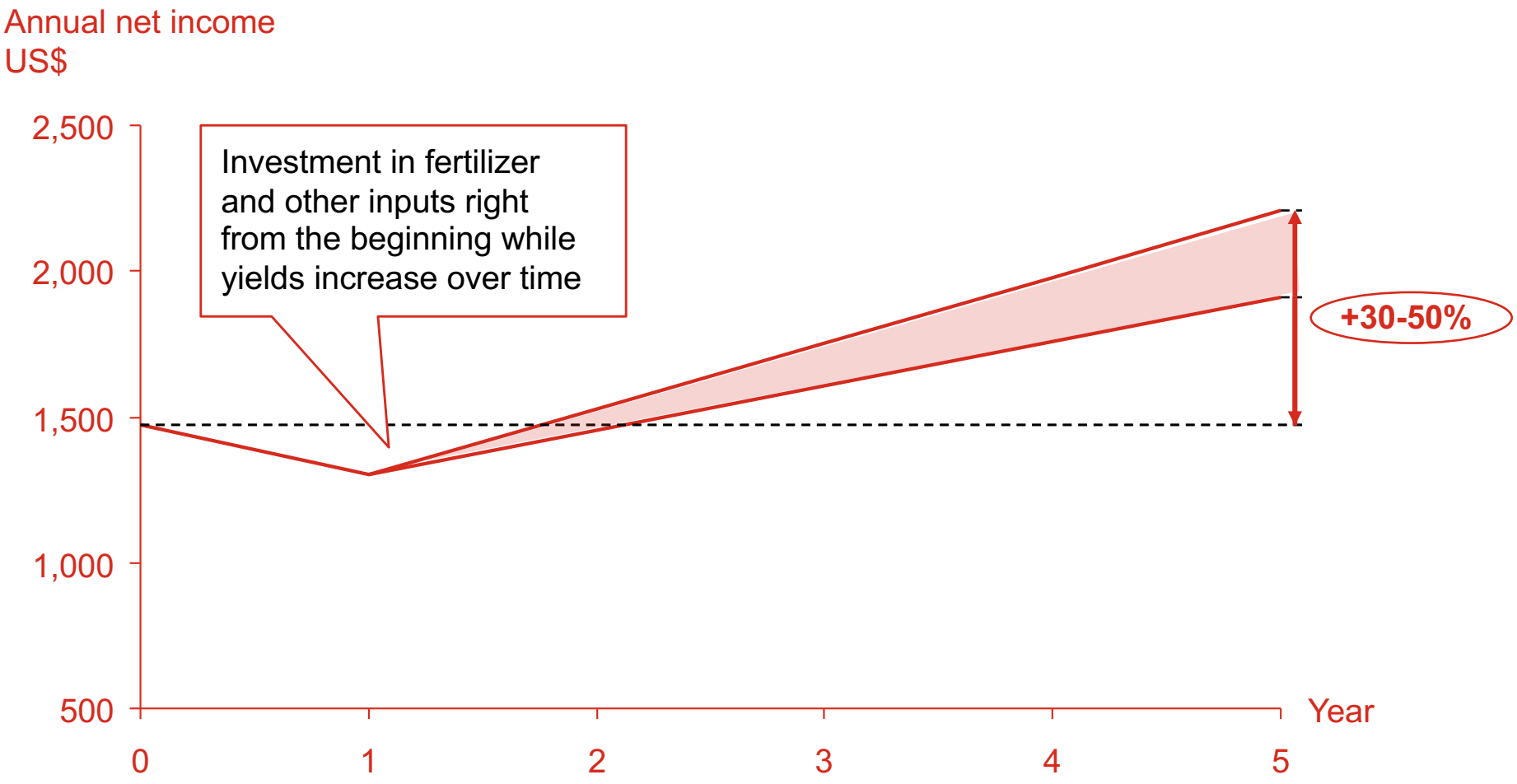
- Increased and optimized fertilization
- Integrated pest management (IPM)
- Control of soil erosion
- Effective shade management
- Proper tree establishment and density

These practices are well-known (and established by Cenicafé), however, the majority of farms have not fully adopted , thus yields are below potential

* According to stakeholder interviews; yield improvement potential depending on current level of adoption of best practices
** Current yield of 930 kg/ha assumes that a significant share of renovation has already happened
Source: Stakeholder interviews

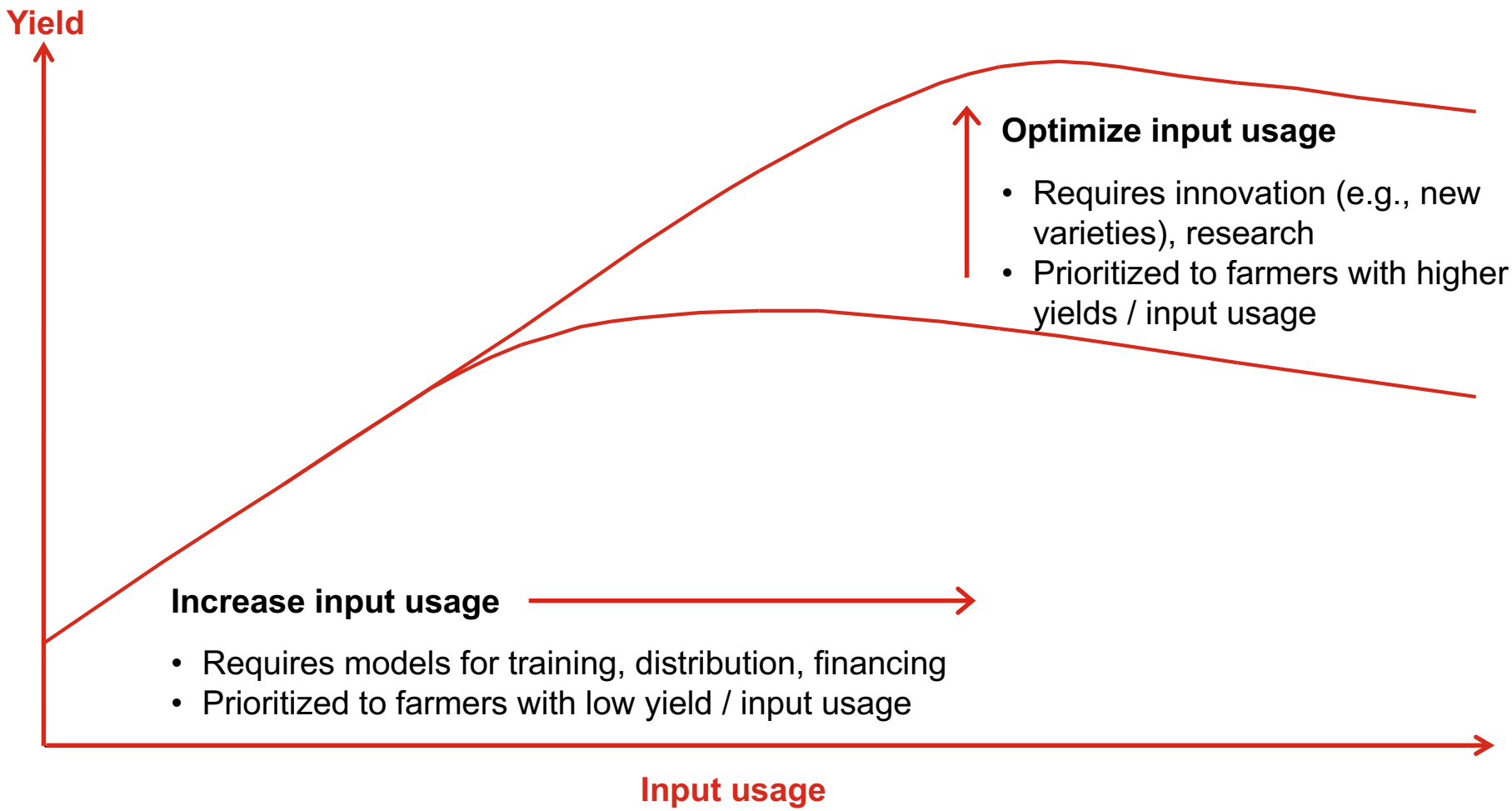
1 Overall, farmer net income would increase by 30 to 50%

Cash-flow for typical smallholder farmer in central region (excluding training costs)*



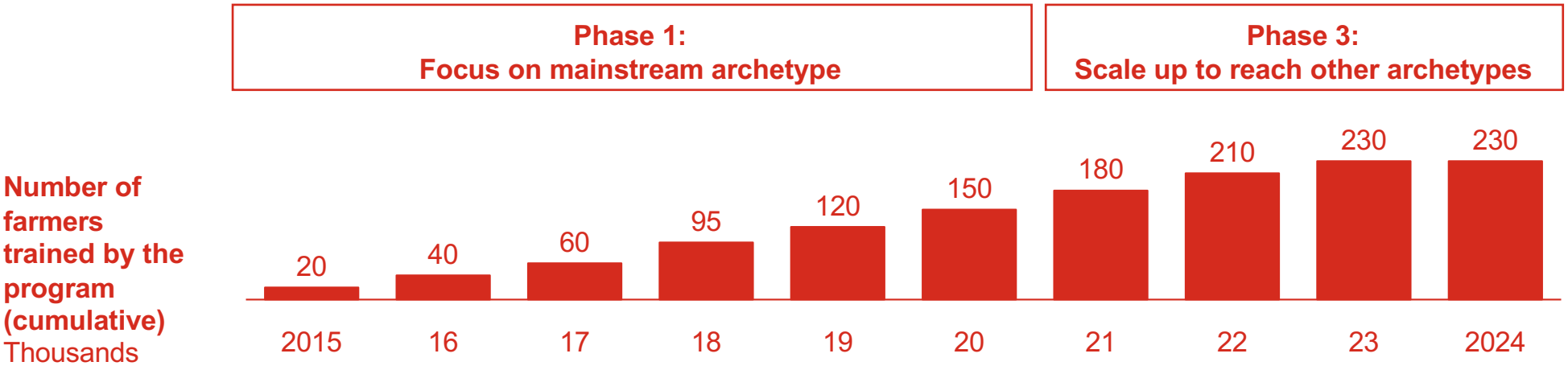
* Assumes farm of 2.0 ha in central region and farm-gate price of $1.50/lb
Source: Interviews with farmers, industry stakeholders and project visits; TechnoServe analysis

1 While some farmers require support to increase input usage, further gains require innovation and input optimization



Source: Adapted from framework developed by Yara

1 An ambitious program to increase adoption could potentially reach up to 50% of Colombia's active coffee farmers



Phase description

- Roll out training program with focus on farmers in mainstream archetypes
- Focus on highest-potential areas and farmer groups at the beginning to pilot and test program
- Scale program to cover 75% of farmers in mainstream archetype

- Scale up program beyond mainstream archetype to include farmers from other archetypes
- Overall, reach ~50% of active farmers

Approx. US$ 200 million* over 10 years

*Total cost of about $800-1,000 per farmer (for two-year program)
Source: TechnoServe analysis

1 Achieving yield improvement requires high adoption level of best practices that might face several challenges

Barriers that might hinder adoption of best practices

High age of coffee farmers

- Increasing average age of coffee farmers as younger – currently 56 years - generations leave rural areas and look for income opportunities in urban areas
- Less likely to adopt new practices and technologies

Lower skill levels

- Lower educational level of many coffee farmers, especially from older generations,
- Might not have the required skill levels to adopt new practices

Remoteness / access

- Difficult to reach farmers that are remotely located and face transport challenges (e.g., bad roads)
- Less likely to ensure adequate coverage to achieve high adoption rates

Source: TechnoServe analysis

1 There are four common types of farmer training models with varying impact and cost

Recommended approach

Training model	How it works	Likely impacts	Cost per farmer
Classroom	1-2 large trainings (50-500 farmers) organized at meeting venue to create awareness and provide basic training	• Effective at teaching a small set of concepts • Typically most useful for preparing for certification	$
Demonstration plots	Demonstration plots established and several trainings (2-4) organized for smaller groups (50-80 farmers) to show impact of GAPs	• Effective at demonstrating GAPs and reaching quick adopters • May be less effective in changing behavior of slower adopters	$$
Farmer field schools	Small demo plots established and frequent trainings (6-12) organized for small groups (20-30) and rigorous data collection and analysis	• Effective at providing hands-on experience for implementation of GAPs • Typically high adoption rate of farmers with varying skill levels	$$$
Individual visits	Individual, personalized training (4-6 visits/year) organized onsite to demonstrate and directly implement GAPs	• Effective for more advanced, technically skilled farmers • May be less effective for farmers who learn better through peer interaction	$$$$

Farmer field schools are recommended based on scalability, impact potential, and the cost of the approach to increase outreach and improve adoption rate of farmer training

Source: MEAS report, stakeholder interviews

1 Innovative strategies to target youth in training program could enhance program's results

Training program should actively involve the youth to ensure that coffee farming continues to be a key source of rural development

Currently many youth migrate to more urban areas as coffee farming does not constitute an interesting income opportunity for many of them

Benefits of including youth on farm-level

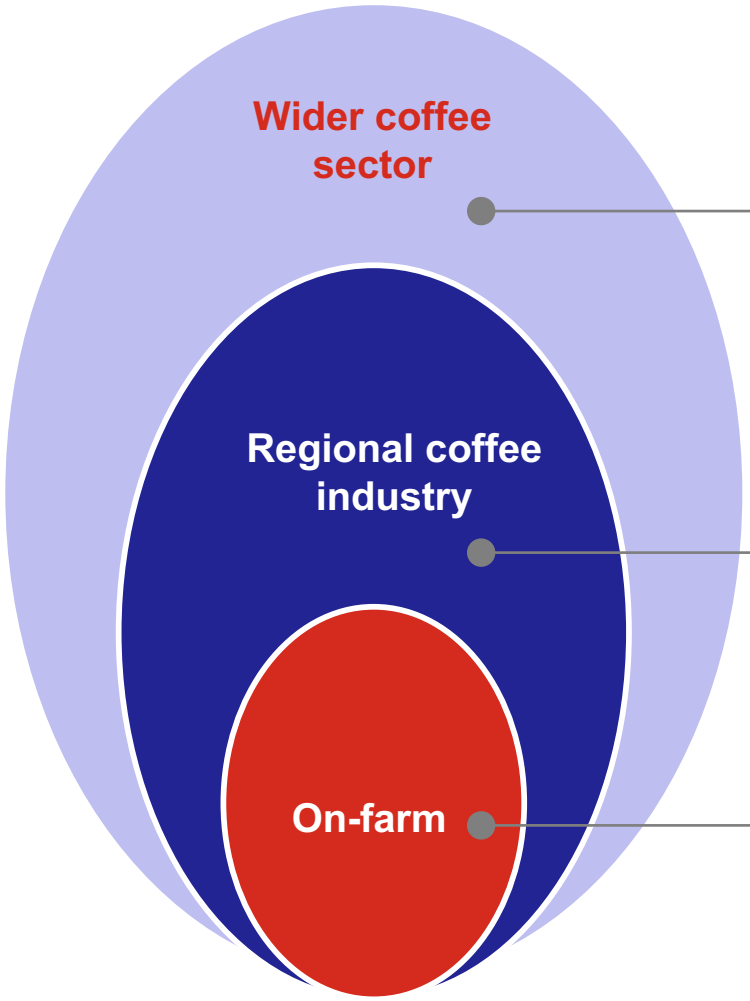
- Higher likelihood of adopting new practices and technology, leading to more productive and efficient coffee farming
- Higher likelihood of managing farm as a business and applying basic business and financial management skills
- More open to invest in coffee farming
- Coffee continues to be a key source for rural employment (both for coffee farmers and coffee workers)

Source: TechnoServe analysis

1 Besides engaging the youth on-farm, the coffee industry offers a variety of employment opportunities

Potential employment opportunities for the youth

NOT EXHAUSTIVE



- Cupper
- Barista
- Fertilizer sales/distribution
- Trainer / auditor for certifier
- Researcher / investigator

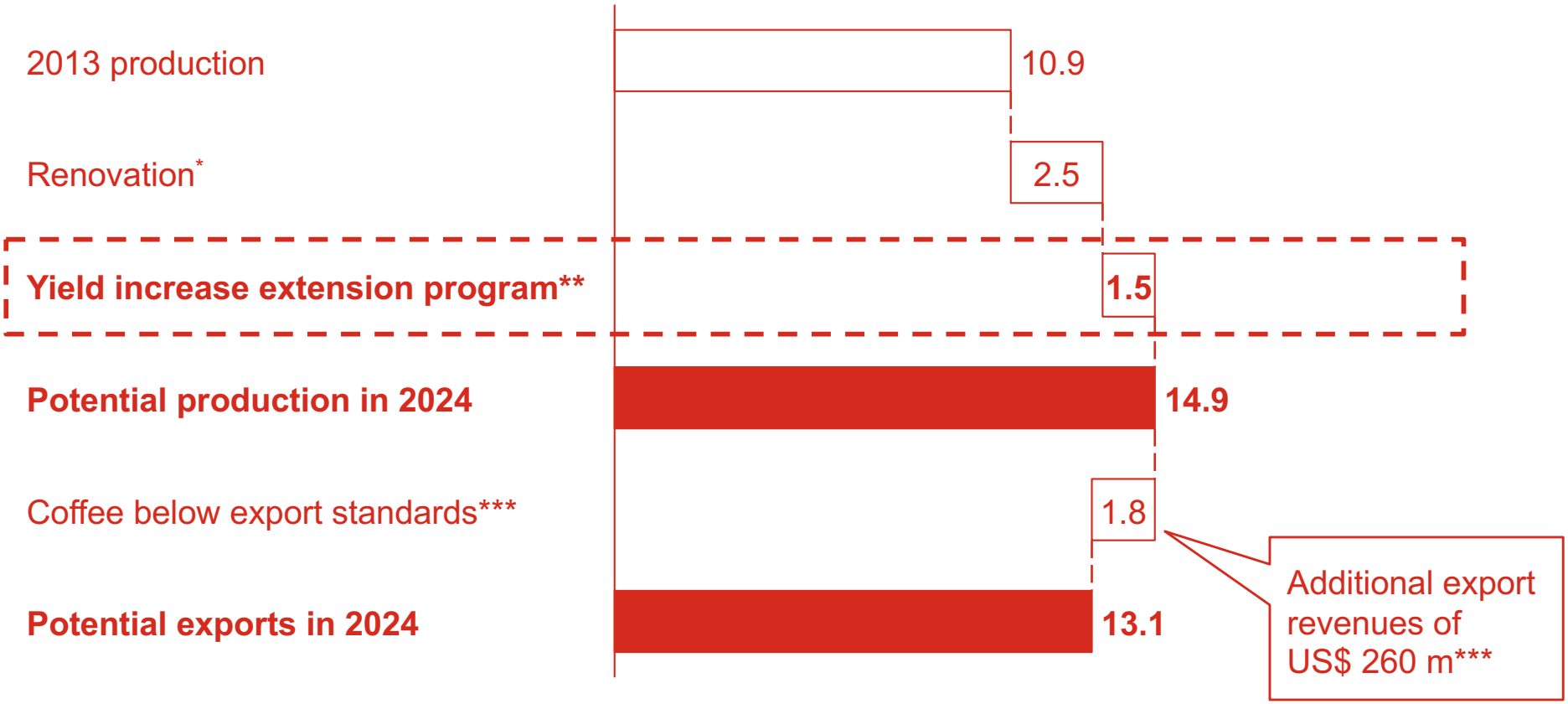
- Truck driver for collecting coffee
- Cooperative management
- Management of central wet mills
- Regional coffee committees

- Farm manager
- Extension worker

Source: TechnoServe analysis

1 This intervention could increase production by at least 1.5 million bags and exports by over US$ 260m after 10 years

Nationwide coffee production by 2024
Million bags green coffee

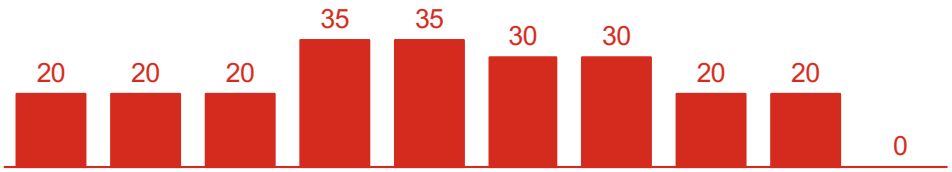


Note: Forecast based on expected yield increase of 20%
* Production from areas already renovated and not in production yet and yield increase from areas to be renovated
** Assuming yield increase will be fully achieved in the 3rd year
*** Assuming 10% of production does not pass export quality standards and price of US$ 1.50 per lb green
Source: FNC; Stakeholder interviews, TechnoServe analysis

1 Overall, the program could generate additional export earnings of US$ 1.3bn over 10 years

The training of farmers can be scaled up over time, reaching 230,000 farmers

Farmers newly enrolled in program, thousands

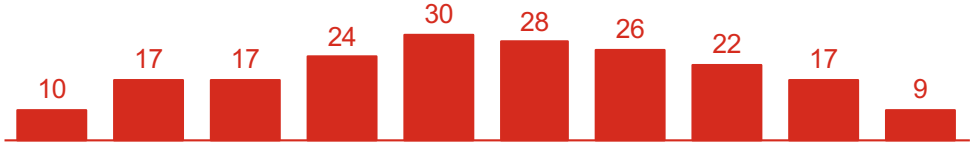


Total over 10 years

230,000 farmers

The training would cost ~US$ 200 m over 10 years

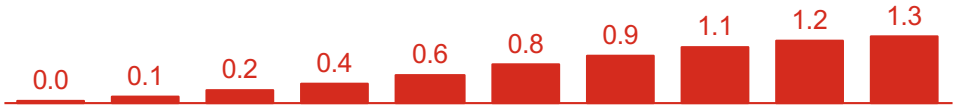
Training costs per year , US$ million



US$ 200 million

Through improving yields, Colombia's exports will increase over time

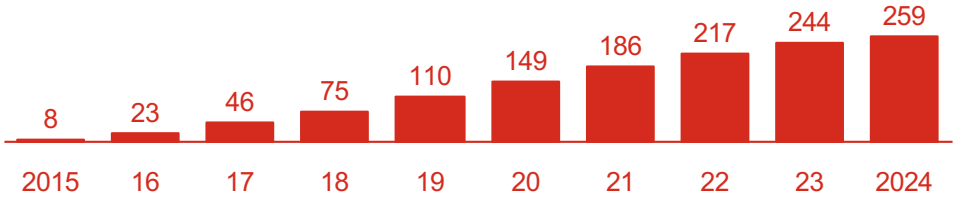
Incremental export volume, million 60kg bags green*



6.6 million bags

Resulting in additional export earnings

Incremental export value, US$ million**



US$ 1.3 bn

* Assuming yield improvement to be achieved over 3 years
** Assuming price of US$ 1.50/lb
Source: TechnoServe analysis

1 Climate change will likely put additional pressure on smallholders, especially on farms at lower altitudes

Climatic variable	Anticipated change	Expected impact
Rising temperatures	• Temperatures are expected to rise by 2.5°C by 2050 • Melting of glaciers in the Andes and disappearance of important moorlands that act as important sources of water	• Decreased groundwater recharge • Decreased water resources • Increased pest and diseases • Higher price volatility
Erratic precipitation	• Average increase in precipitation of 2.5% by 2050 (3-5% expected for coffee growing areas) • Driest periods will likely be less dry, while wettest periods are projected to become wetter	• Increased risk of soil erosion and suboptimal infiltration of saturated soils • More unfavorable conditions for drying of coffee

Farms at lower altitudes are most vulnerable to impact of climate change and should be targeted for diversification opportunities

Source: CIAT

BUSINESS CASE DISCUSSION

Business case 2: Improving sector efficiencies

Business cases and key challenges

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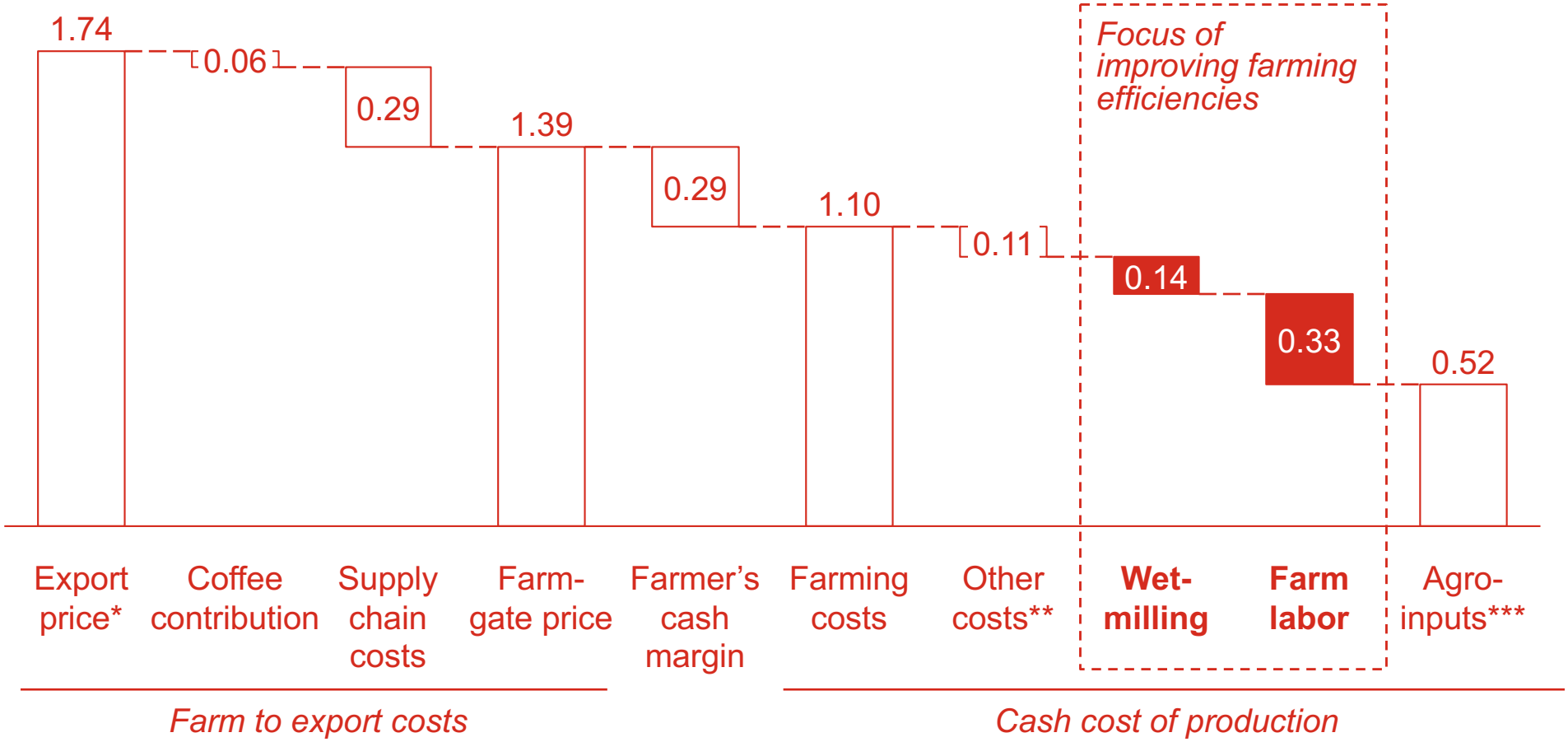
2 There may be modest opportunities to reduce the cost of production in Colombia's supply chain

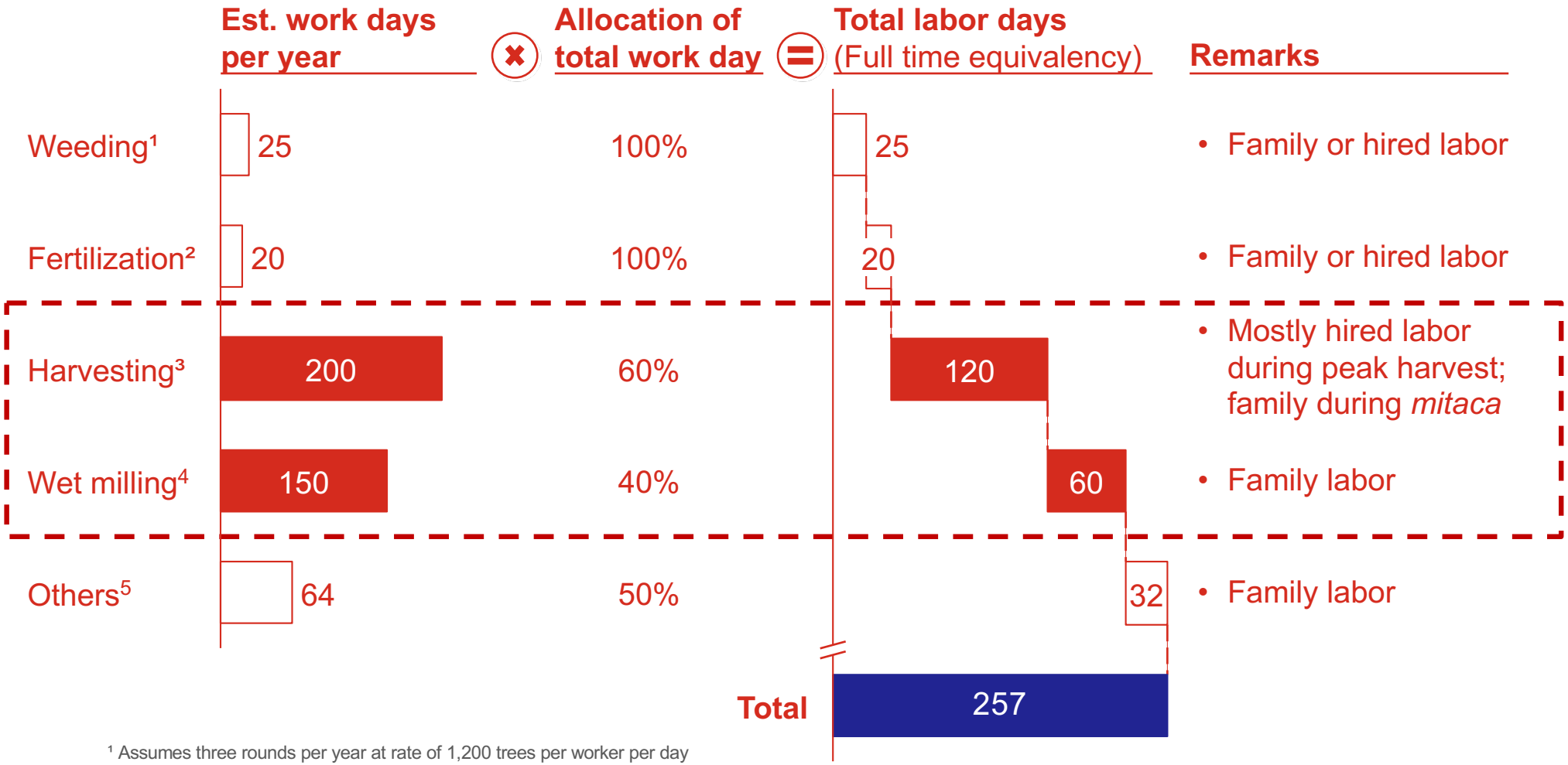
Cash cost of production for average Colombian farm
US$ per lb green



Note: Cost of production does not include costs for installation of wet milling equipment, tree renovation or financing

*Normalized to 2012/13 average commodity prices – Arabica ICE "C" of $1.50 but adjusted for market differentials)
** Includes farm tools, equipment, internal transport, etc.
*** May be possible through optimized fertilizer and distribution methods (see Business Case 1, page 47)
Source: Stakeholder interviews, TechnoServe analysis

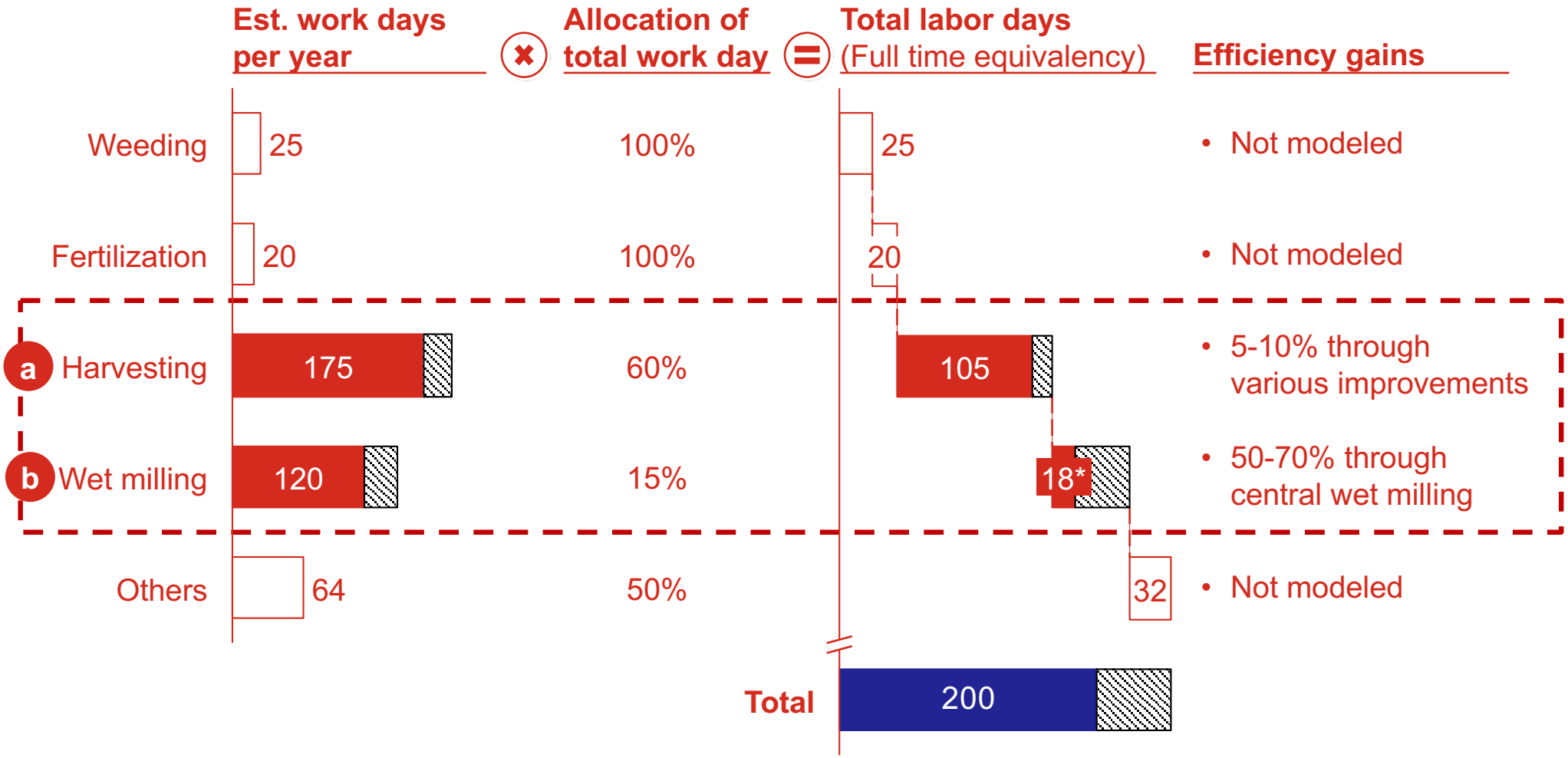
2 Harvesting and home wet-milling comprise 70% of total on-farm labor requirements



[1] Assumes three rounds per year at rate of 1,200 trees per worker per day
[2] Assumes two rounds per year at rate of 1,000 trees per worker per day
[3] During main crop, assumes 80 kg/day (cherry) x 1.5 workers x 60 harvest days; during *mitaca*, assumes 35 kg/day x 0.5 workers x 60 harvest days
[4] During main crop, assumes 60 days of pulping/washing/drying at 5 hours/day, plus 15 days after harvest for final drying and transport at 2 hours/day; during *mitaca*, assumes 60 days of pulping/washing/drying at 2.25 hours/day, plus 15 days after harvest for final drying and transport at 1 hour/day
[5] Includes replanting, pest control, shade management, erosion control, and transport

NOTE: Modeled on 2 ha farm with 5,000 trees per ha and yield of 850 kg/ha green
Source: CRECE, interviews, TechnoServe analysis

2 There may be ways to reduce farmers' labor requirements in harvesting and wet-milling, saving them time and some money



* Home wet-milling time replaced with cherry transport time; during main crop, assumes 1.5 hours for 60 days; during *mitaca*, assumes 1 hour for 60 days

NOTE: Modeled on 2 ha farm with 5,000 trees per ha and yield of 850 kg/ha green
Source: CRECE, interviews, TechnoServe analysis

2a While conceptually there may be ways to reduce harvesting costs, implementation will be difficult

Improvement opportunity	Description	Implementation challenges
Faster picking	• Train farm workers on more efficient picking practices to pick cherries faster • Could potentially involve the use of support tools	• Mostly external labor used for harvesting • High fluctuation of external farm labor • Farm laborers mostly receive variable payments based on cherry volumes picked • Risk to lower quality when improving picking speed
Mechanical harvesting	• Use machines to improve harvesting efficiency and lower manual labor requirement	• Topography - relatively flat landscape required to use machines • Harvesting throughout the year • Risk to lower coffee bean quality
Leveled terrain	• Terrace coffee farms to establish flat coffee areas that allow to improve picking speed and potentially use harvesting machines	• Expensive • Topography is difficult

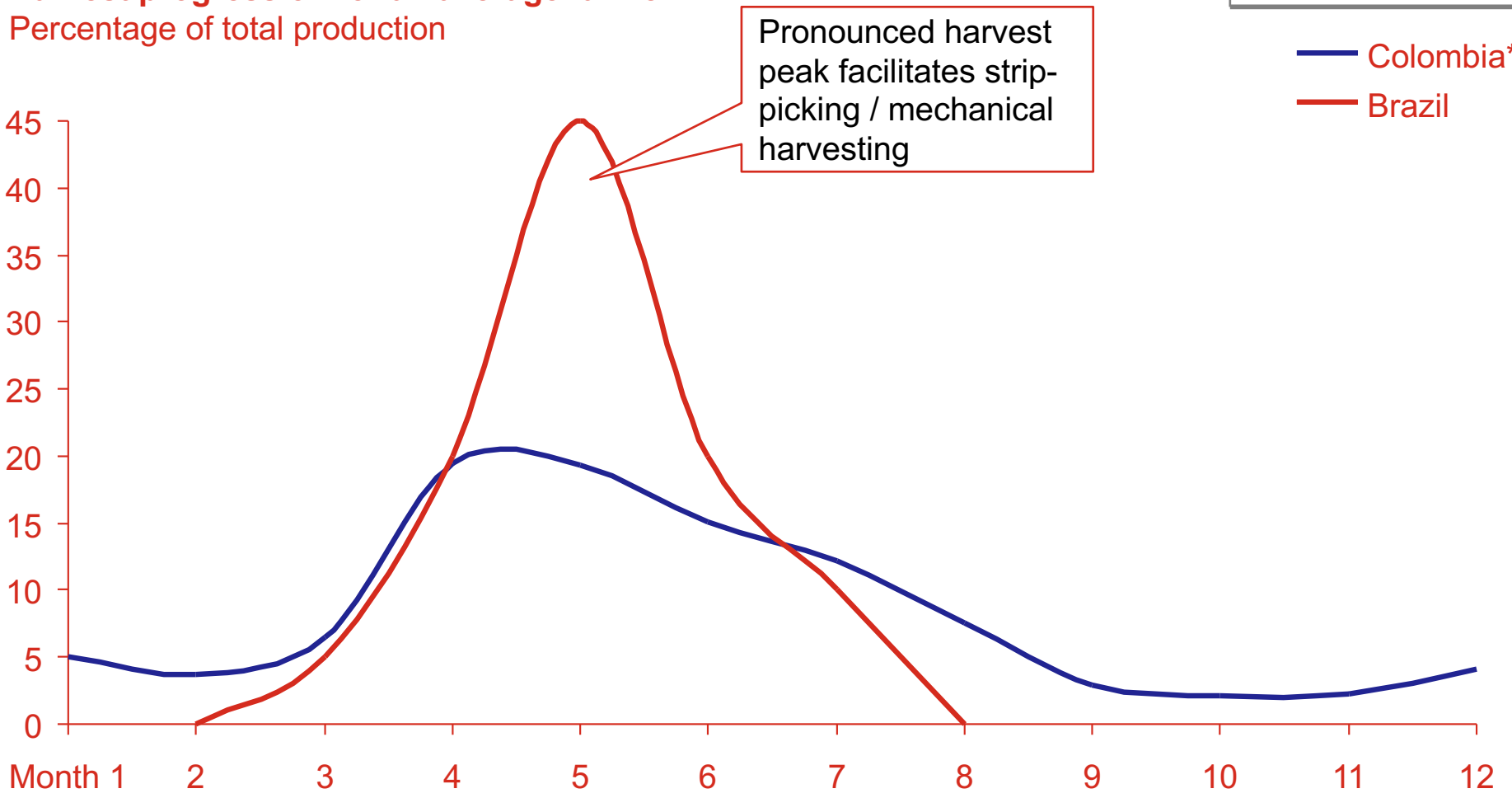
Critical to assess whether cost savings potential justify implementation costs for achieving increased harvesting productivity

Source: TechnoServe analysis

2a Due to agro-climatic factors, there is limited opportunity to apply technology for improving harvesting productivity

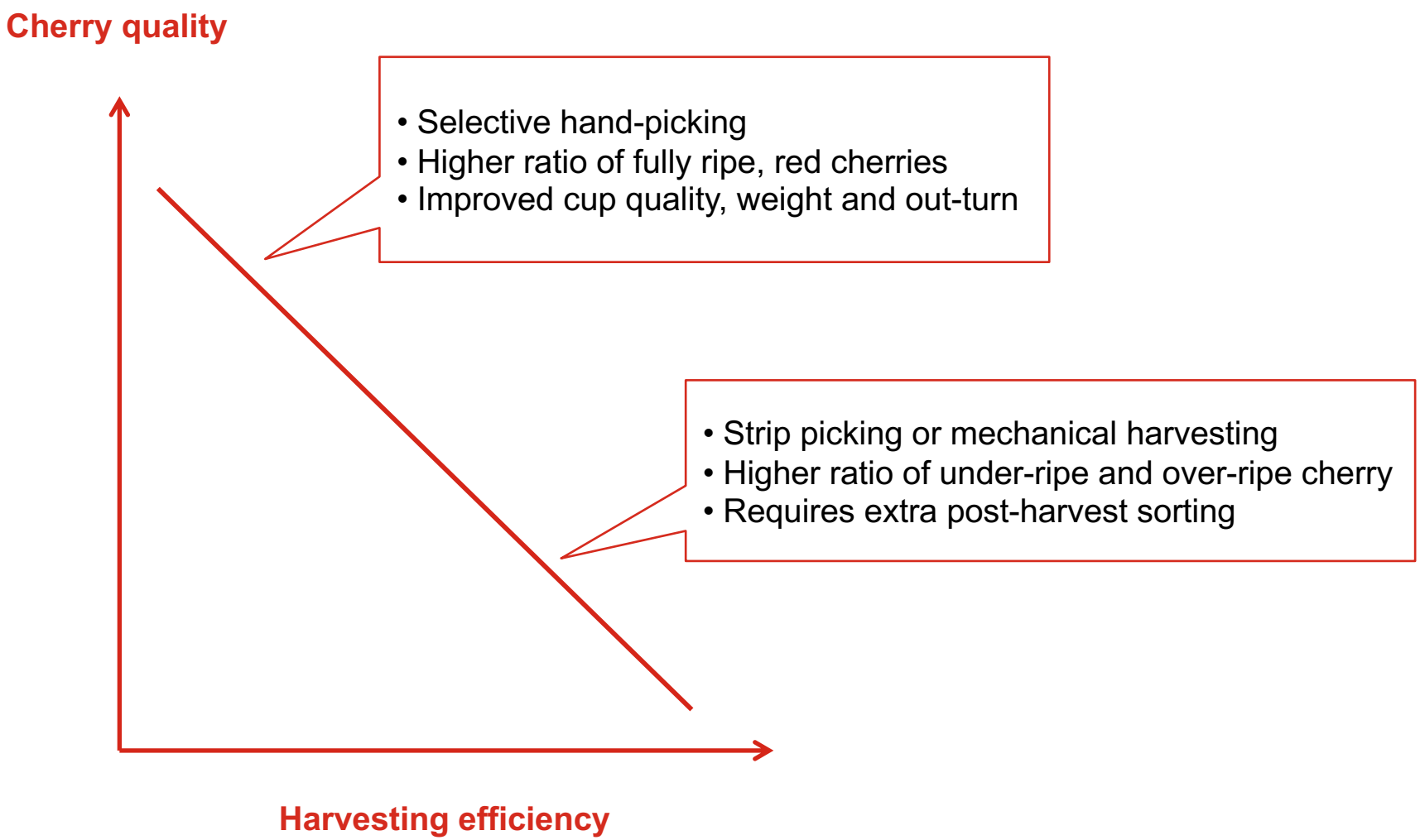
COLOMBIA DATA BASED ON HUILA EXAMPLE

Harvest progression for an average farmer
Percentage of total production



* Based on data from Huila
Source: Stakeholder interviews, TechnoServe analysis

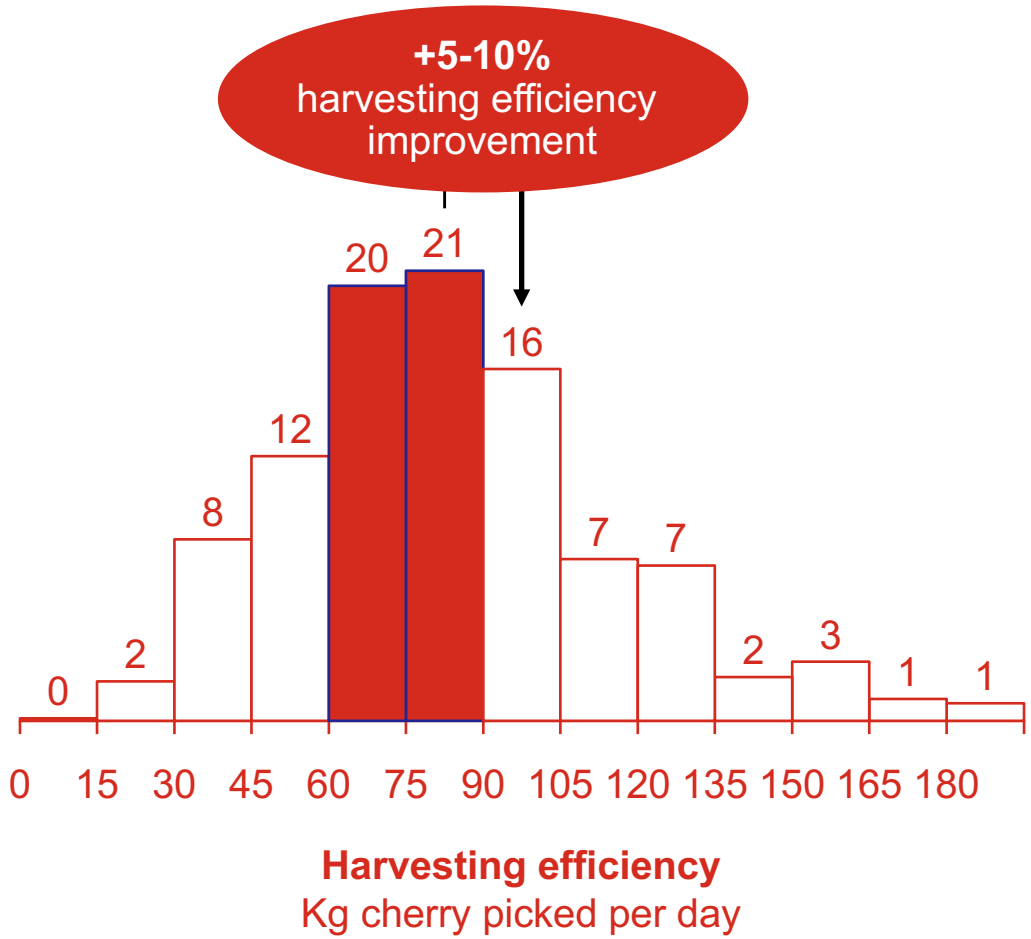
2a There is usually a trade-off between quality and harvesting efficiency



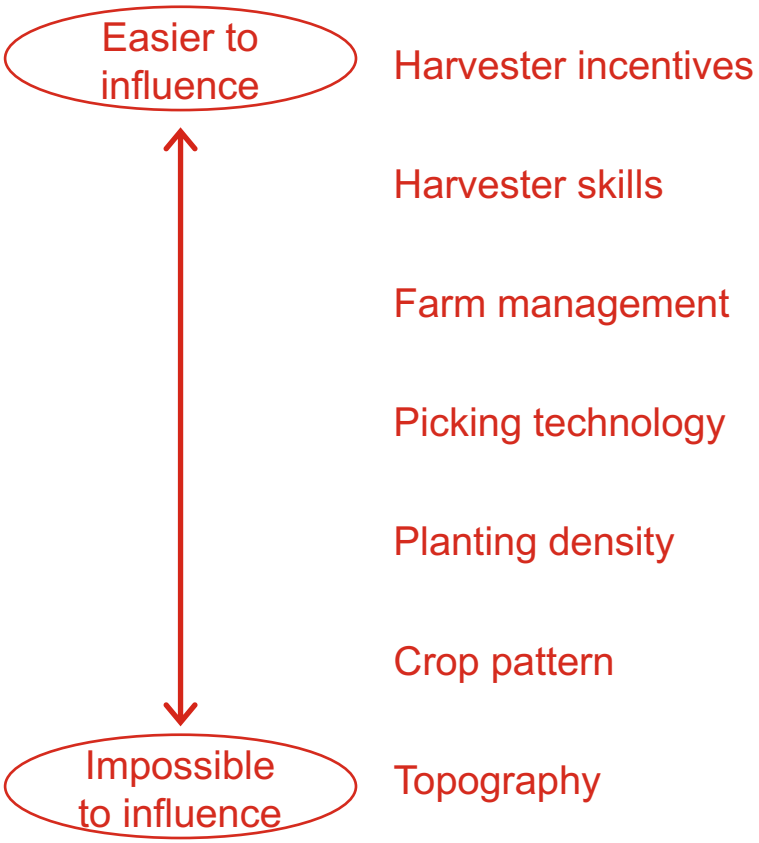
Source: TechnoServe analysis

2a A combination of approaches could likely deliver a 5 - 10% harvesting labor efficiency improvement over time

Frequency distribution of harvesting labor productivity
% of surveyed harvest workers

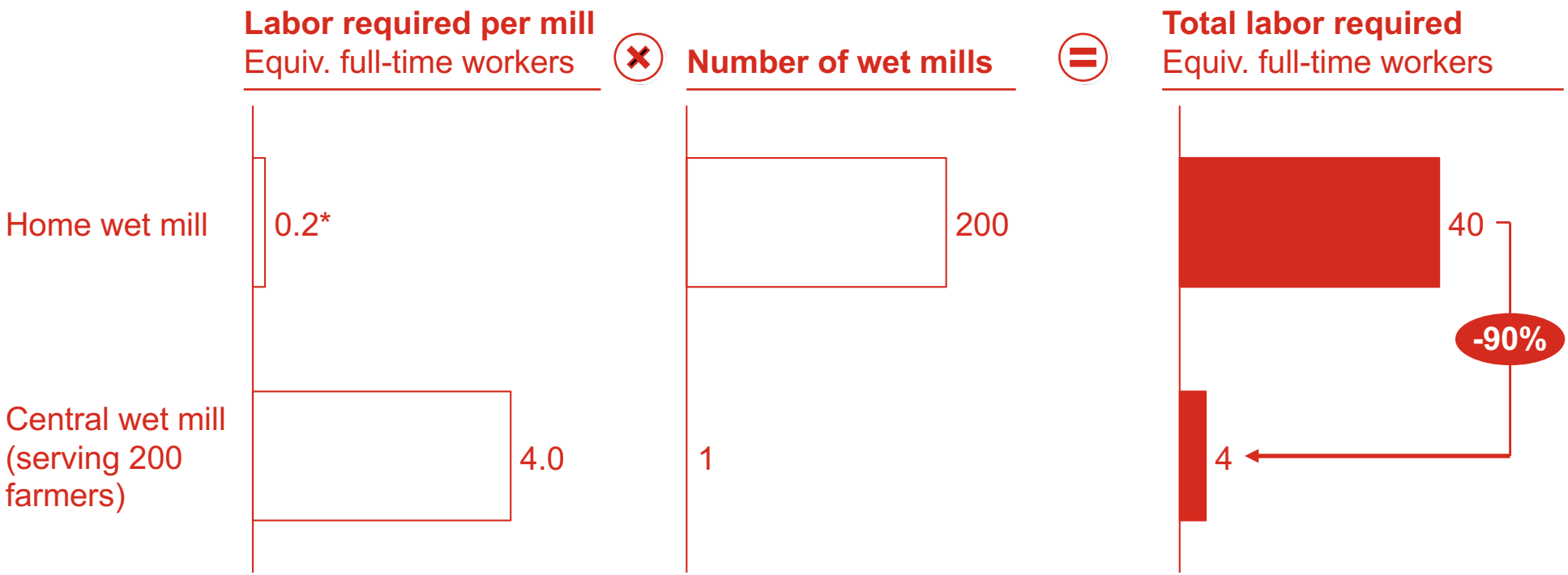


Factors influencing harvesting labor productivity



Source: CENICAFÉ; based on a study conducted in Manizales, Chinchiná, Palestina and Risaralda, TechnoServe analysis

2b Centralized wet milling could reduce the need for wet mill labor by 90% although it is mostly family labor



- Central wet milling leads to more than 90% labor time savings
- However, it is difficult to value this saving potential as the majority of home wet-milling labor is family (unpaid)

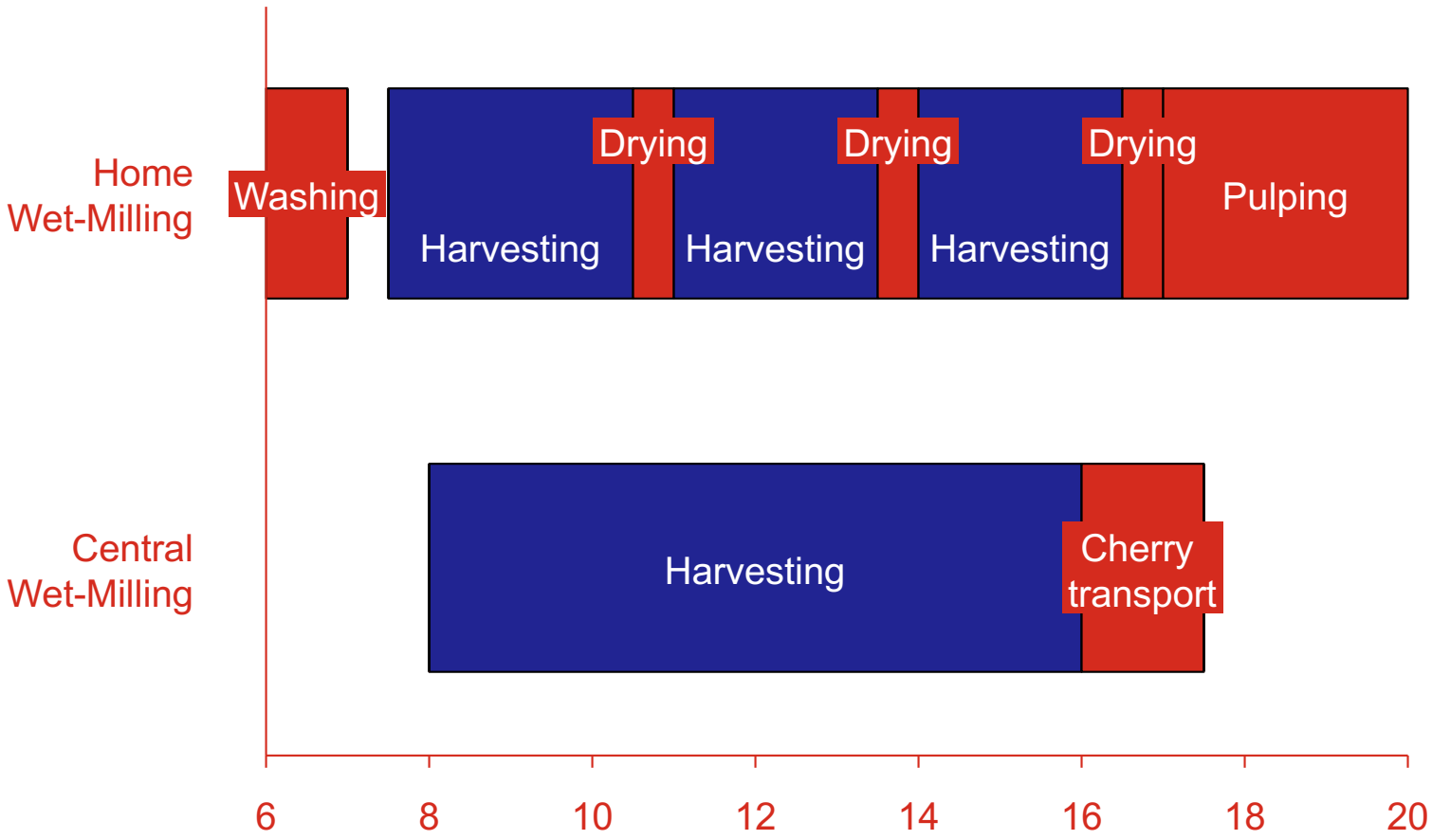
* Assuming 60 out of 300 working days per year (including Saturdays)
Source: Stakeholder interviews, TechnoServe analysis

2b Farmers would save 3-4 hours per day during the main harvest period

ILLUSTRATIVE

Time allocation for coffee farming family during peak harvest

Total time spent on wet-milling



5 hrs / day

1.5 hrs / day

Source: TechnoServe analysis

2b However, it is uncertain how farmers will value or invest this additional time

Options how farmers could use additional time		Result
Substitute paid labor	• Farmers use additional time to take over some of the harvesting work previously done by external labor	Labor cash cost savings
Generate additional income	• Farmers use additional time to pursue income generating opportunities, e.g., casual work, small business, growing other crops, etc. • Income opportunities will vary considerably by locality	Higher overall household income
Do not use additional time	• Farmers use additional time for leisure, family activities, etc., rather than investing it in additional work	Social benefits (higher quality of life)

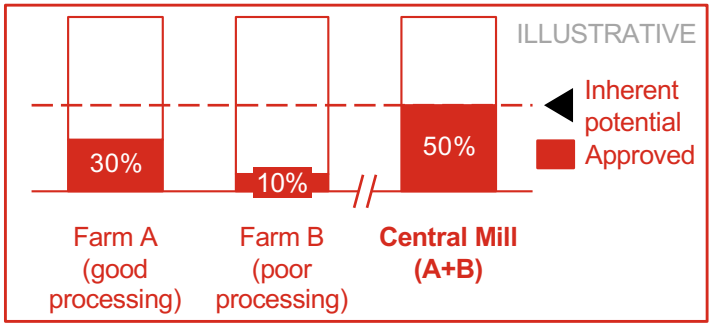
More research is necessary to understand how farmers value their time and respond to the creation of additional time

Source: TechnoServe analysis

2b Besides improving labor productivity, central wet mills provide additional benefits

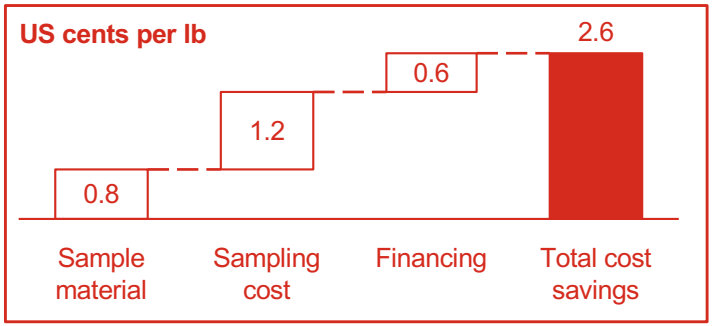
Improved quality and consistency

- **Improved quality** by reducing processing-related defects that are common at home wet-milling, e.g., over-fermentation, poor drying / storage, etc.
- **Improved consistency** by employing standardized quality controls, versus each farmer employing different practices at his/her home wet mill



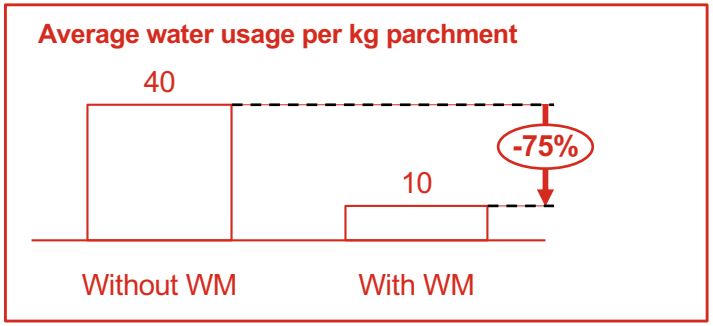
Reduced aggregation costs

- **Reduced sampling losses and costs** by eliminating the need for a parchment sample to be taken from each delivery from each farmer in order to assess cup quality and net green weight (factor)
- **Faster payment** for farmer by paying upon delivery of cherry instead of parchment (savings of 7-10 days and improved cash flow)



Improved environmental compliance

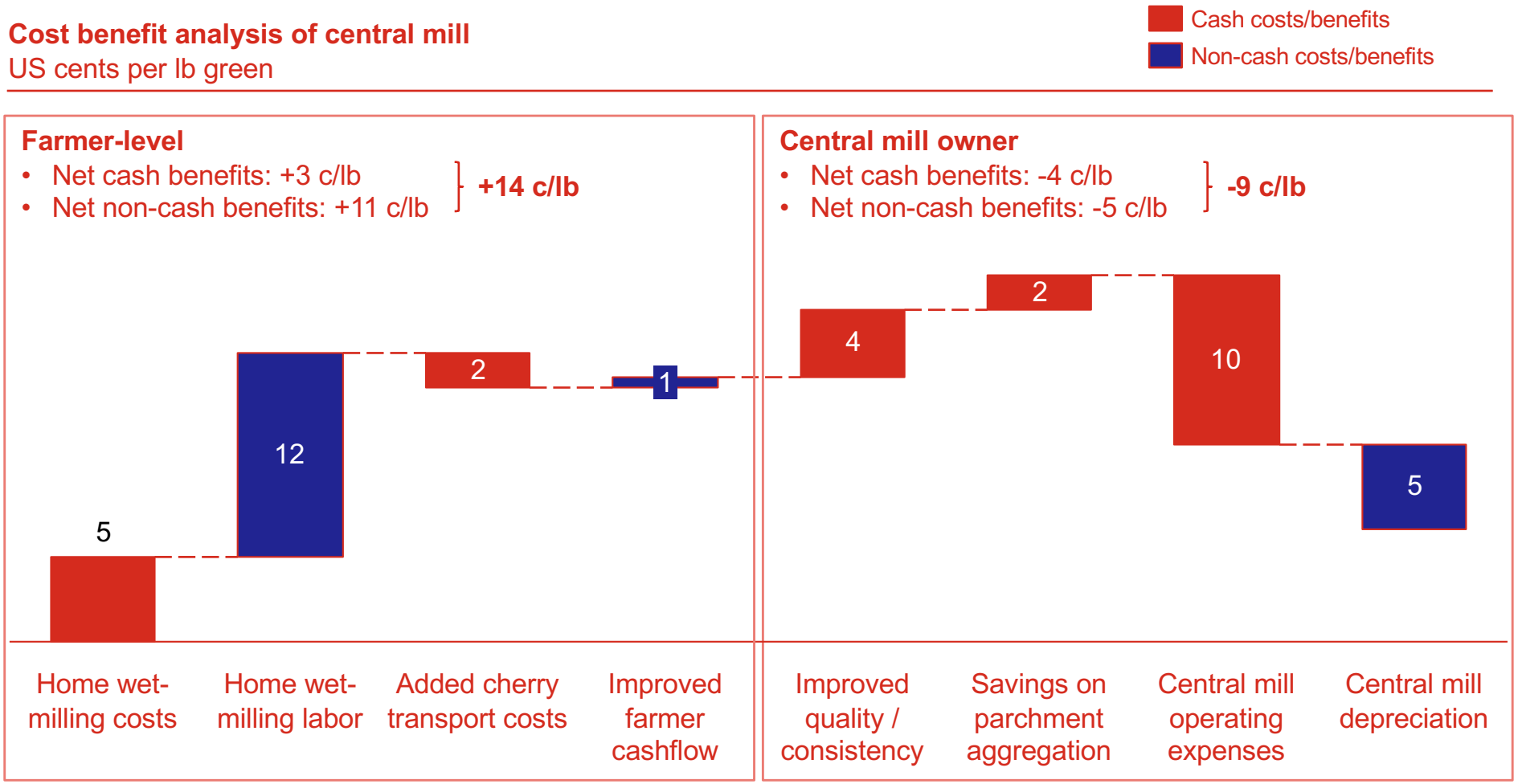
- **Significantly lower water usage** through eco-pulping technology
- **Simplified wastewater treatment** by installing a system for one mill, instead of for every farm
- **Improved and more cost-effective compliance with laws and certification** standards



Source: Stakeholder interviews, TechnoServe analysis

2b These benefits are likely to be more attractive to farmers than to potential investors

Cost benefit analysis of central mill
US cents per lb green



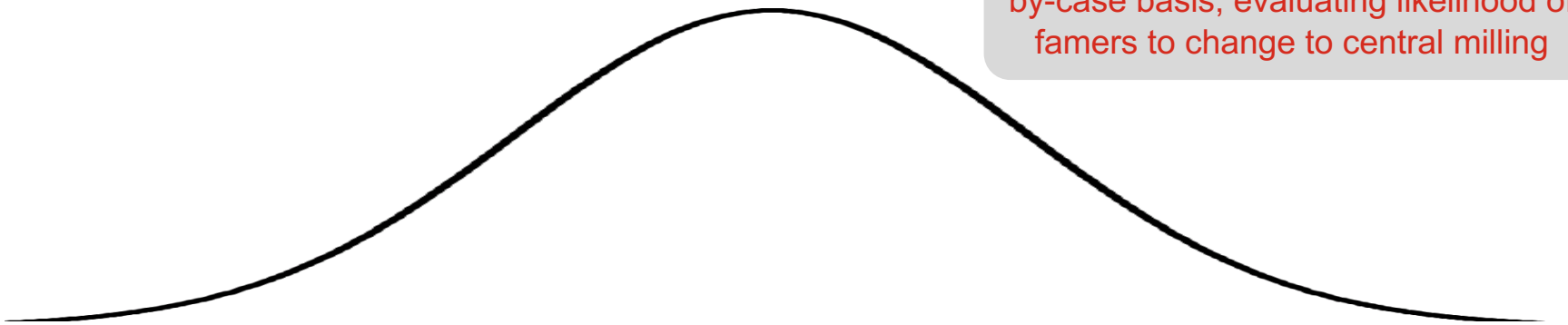
The overall cash benefit is negative (-1 c/lb), but the overall non-cash benefit is positive (+6 c/lb). Thus, central mills do create some added cost for the overall sector, but this is justified by reducing the amount of time required from farmers.

Source: TechnoServe analysis

2b Some farmers are likely to shift from selling parchment to selling cherry more quickly than others

Rate of adoption of central wet mills

Viability should be assessed on a case-by-case basis, evaluating likelihood of famers to change to central milling



	Fast adopters	Slow adopters
Farm traits	• Insufficient home wet-milling facilities (e.g., lack of drying space) • High opportunity cost of labor / time • High potential to improve quality • Committed buyer present • Good road access	• Recently built and well-maintained home wet-milling facilities • Low opportunity cost of labor / time • Less potential to improve quality • No committed buyer • Poor road access / remote areas
	Particularly relevant to archetypes in higher altitudes in the South (lack of infrastructure) and central parts with high opportunity costs	Particularly relevant to archetypes in lower altitudes in the South with low opportunity costs and in remote areas

Source: Stakeholder interviews, TechnoServe analysis

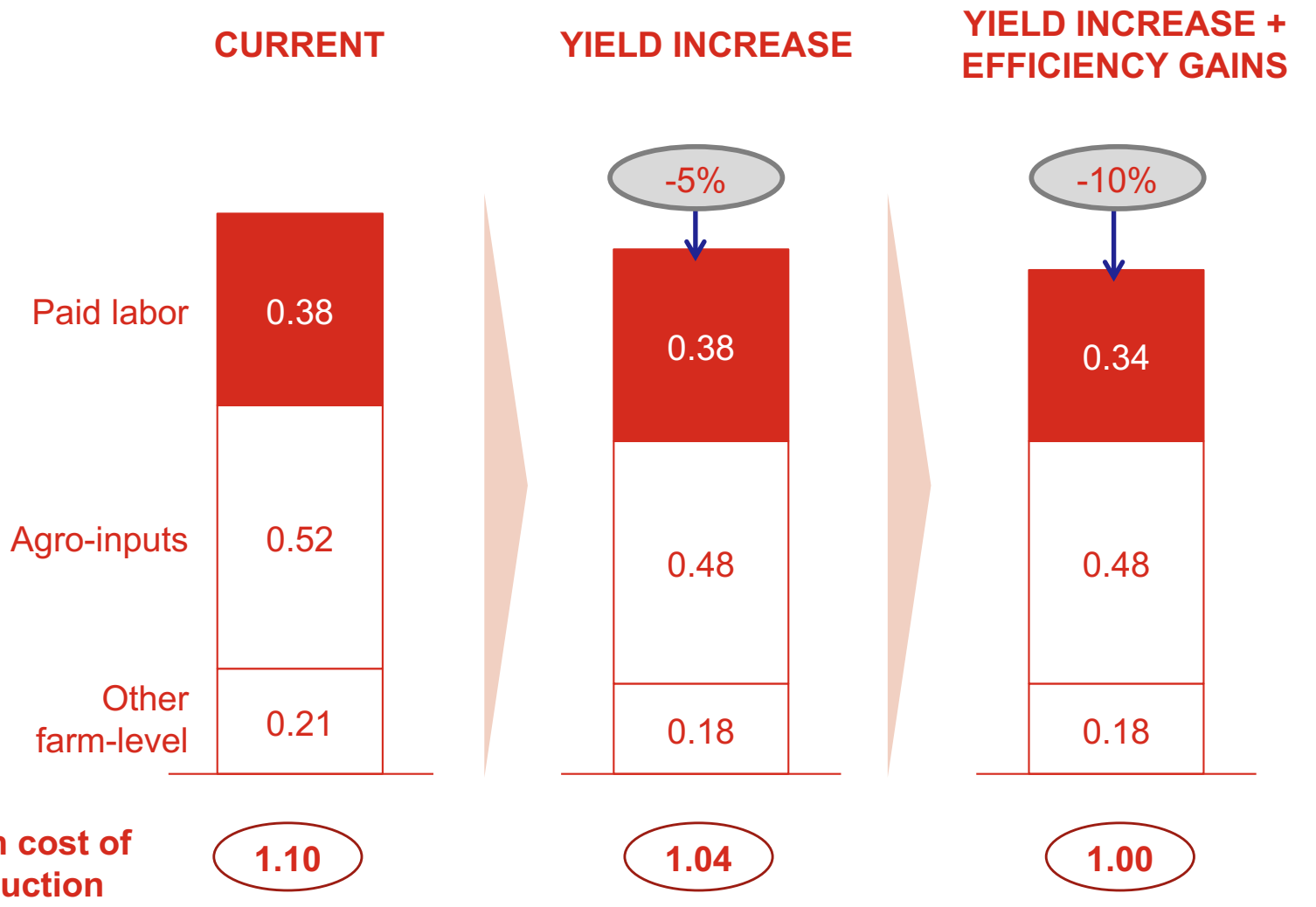
2b Several risks have to be managed to ensure successful implementation of central wet mills

Risk	Description	Mitigating factors
Reduced negotiating power for farmers	Unprocessed coffee needs to be sold quicker and might lead to reduced negotiating power for farmers as they need to sell cherries within a specific timeframe. Farmers might not be able to hold coffee to wait for better prices	• Ensure balanced ownership structure and competition • Ensure transparency • Regular monitoring of purchase prices
Ownership and corruption	Potential for corruption in management of wet mill (misusage of wet mill income; farmers do not receive fair prices)	• Establish sound financing model, using a mix of local banks and international investors • Implement sound management controls • Implement rigid evaluation and monitoring system • Provide training on governance
Negative environmental impact	Wet mills can have negative environmental impact as pollution is highly concentrated in wastewater	• Ensure correct management of concentrated wastewater pollution through implementation of adequate systems/ processes
Low capacity usage	Farmers might be reluctant to sell their whole harvest to wet mill, e.g., due to high transport cost	• Inform farmers on costs and benefits of using wet mill • Assess transport aspect on a case by case basis and try to work with local/ regional authorities to overcome potential barriers

Source: Stakeholder interviews, TechnoServe analysis

2 Overall, on farm costs could be reduced by up to 10%

Average on-farm costs for mainstream archetypes*
US$ per lb green



* Including cash and non-cash costs
Source: Interviews; TechnoServe analysis

BUSINESS CASE DISCUSSION

Business case 3: Improving environmental management

Business cases and key challenges

1 Farmers can increase yields through improved agronomy and technical assistance

- Colombia's farm yields could be increased by 20 to 30% by transferring improved agronomy practices to farmers, in particular effective fertilizer usage
- Achieving yield improvement requires high adoption level of best practices; high average age of farmer population could pose a challenge
- Higher yields contribute to a 5% cost reduction and net income increase of 30-50%
- Overall, production would increase by 1.5M bags and exports by US$ 260M per year
- Cost of US$ 200M to train 50% of active farmers (230,000) over 10 years
- Training should complement FNC's individual farm extension visit s with a "farmer field schools" methodology and should actively try to involve the youth

2 Production costs could be reduced modestly through various sector efficiencies

- Colombia's coffee farming is already quite sophisticated and shows limited potential for reducing production costs
- By reducing farmers' labor requirements for harvesting and in particular home wet-milling, production costs could be reduced by up to 10%,
- For harvesting, there are conceptually opportunities for improving productivity; however, implementation will be difficult and results may not justify costs
- For wet-milling, the introduction of central milling could reduce labor requirements by 50-70% and produce other financial and nonfinancial benefits
- Farmers facing the highest opportunity costs to time and labor should be prioritized

3 Supply chains can meet stricter environmental standards with minimal added cost

- Challenging for farms to meet new, stricter environmental regulations – requires new wastewater treatment systems that are complicated and expensive
- Ensure compliance with environmental laws cost-effectively through central wet mills; a single system now benefits many farms

- *The combination of all three business cases would increase net income by more than 50% and reduce time spent for coffee farmers, enabling them to earn more from coffee than other rural employment options*